Frameworx Reference

Core Frameworks Concepts and Principles

Business Process, Information and Application Frameworks

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Notice

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Executive Summary

TM Forum Frameworx is a suite of best practices and standards that provides the blueprint for effective, efficient business operations and enable a service-oriented, highly automated and efficient approach to business operations and application integration.

This Concepts and Principles document strives to enable the reader to get a complete point of view of the compilation and creation of the core Frameworks; (business process, Information and application frameworks), explaining the reasoning to its construction.

Frameworx Design Principles

Frameworx is built on a services oriented design and uses standard, reusable, generic blocks that can be assembled in unique ways to gain the advantages of standardization while still allowing customization and enabling differentiation and competition at the service level.

The components that make up the TM Forum Frameworx include the following:

1. The Business Process Framework (formerly known as eTOM)
2. The Information Framework (formerly known as SID)
3. The Application Framework (formerly known as TAM)
4. Business Metrics
5. Best Practices

This guidebook focuses on the top three core elements of Frameworx from the above list: The Business Process, Information and Application frameworks, called the core Frameworks.

All core frameworks are organized around domains; the definition of a domain is as follows:

A domain represents a unit correlated to a specific management area.

In the Business process framework it comprises distinct groupings of processes pertinent to a specific management area.

In the Information framework it comprises a collection of (aggregate business entities) whose lifecycle is the responsibility of the management area.

In the Application Framework it comprises a set of applications that are specific to the management area.
Domains provide a common architectural construct for processes, information and applications and have the following properties:

- *Contain processes, business entities, applications, and interfaces that encapsulate both operations/behavior and corporate/enterprise information for the domain's business objectives*
- *Represent a distinguishable share of the enterprise's operations*

They are closely related (very cohesive) collections of corporate/enterprise processes, business entities, applications, and interfaces to fulfill the domain's role within the enterprise.

Nota bene: The Business Process Framework and the Information Framework (Product & Service domains) were approved as official ITU-T standards (M-3050 and M-3190 respectively) in 2004 and 2006. See References for further information on ITU-T TMN and 'M' Series.

The Business Process Framework

The purpose of the Business Process Framework is to continue to set a vision for the CSP industry to compete successfully through the implementation of business process driven approaches to managing the enterprise. This includes ensuring integration among all vital enterprise support systems concerned with service delivery and support.

The focus of the Business Process Framework is on the business processes used by service providers, the linkages between these processes, and the use of Market, Product, Customer, Service, Resource, Engaged Party, and Enterprise Management related information by multiple processes.

The over-arching objective of the Business Process Framework is to continue to build on TM Forum's success in establishing:

- An (CSP) 'industry standard' business process framework.
- Common definitions to describe process elements of a CSP.
- Agreement on the basic information required to perform each process element within a business activity, and use of this within the overall Frameworx program for business requirements and information model development that can guide industry agreement on contract interfaces, shared data model elements, and supporting system infrastructure and products.

The Information Framework

The Information Framework provides an information reference model and a common information data vocabulary from a business entity perspective.

Like the other Core Frameworks components, the Information Framework uses the concepts of domains and aggregate business entities (or sub-domains) to categorize business entities, so as to reduce duplication and overlap. Based on data affinity concepts, the categorization scheme is necessarily layered, with each layer identifying in more detail the "things" associated with the immediate parent layer. This partitioning of the Information Framework also allows distributed work teams to build out the model definitions while minimizing the flow-on impacts across the model.

The Information Framework provides CSPs with and a data entity view of their business. That is to say, the Information Framework provides the definition of the 'things' that are to be affected by the business processes defined in the Business Process Framework.

The Application Framework

At its core, the Application Framework is a logical application map supporting the business processes in a Communications Service Provider type of enterprise. It includes descriptions and hierarchal groupings that are structured based on the common horizontal domains in Core Frameworks.

The Application Framework provides its own decomposition and methods for identifying, classifying and grouping of concepts. The identification guidelines include commercial and best IT practices.

Reasoning for grouping of applications can vary. The document suggests a number of cases for grouping throughout the Application Framework. These can be based on applications that consume other application's functionality or share functionality with another application. Grouping can also be based on similar end users, purpose, or data entity.

The classification system in the Application Framework makes reuse of the Business Process and Information Framework decompositions and the domains in Frameworx, along with its own methods to specify integration infrastructure functionality.

PART I – FRAMEWORX COMPONENTS - STATUS & RELATIONSHIPS

1 Purpose

The purpose of this section on core Frameworks is to outline the relationship between the processes as defined in the Business Process Framework, the information elements as defined in the Information Framework and the Application Framework specification of applications and their associated functionality and other characteristics.

2 Core Framework Component Status definition

As core Frameworks Components evolve, they go through the following evolutionary stages or statuses:

- Released
- Preliminary
- Draft
- Not Fully Developed

2.1 Released

A core Frameworks Component with a Released Status is:

- Complete, i.e. no additions required
- Frameworx Team (maturity level 3) and Forum approved (level 4)

A core Frameworks Component with a Released Status is included as the basis for TM Forum Frameworx Conformance Certification.

2.2 Preliminary

Preliminary is typical work that is delivered over a series of iterations.

A core Frameworks Component with a Preliminary Status is:

- Incomplete, i.e. a deeper study is in-progress to complete the content of the item and it is usable as it is. Some modification might be done.
    - Frameworx Team (maturity level 3) and Forum approved (level 4)

Note that even if a Preliminary component is approved by the Frameworx Team and the Forum, it won't be included in TM Forum Conformance Certification as it has to be completed.

2.3 Draft

A Core Frameworks Component with a Draft Status is:

- Incomplete, i.e. a deeper study is in-progress to complete the content of the item and present it to approval
- Not yet Frameworx Team and Forum approved (maturity level 1 or 2)

Note: A Draft work can't remove or move any existing item and all new items have to be gathered in a new area with a Draft maturity level

2.4 Not Fully Developed

A core Frameworks Component with a not fully developed status is

- Empty or almost empty, i.e. no work has started to specify it but it has been identified as required

In case of almost empty the Frameworx Team has to review the item and decide if it has to be removed.

2.5 How to specify the Status?

2.5.1 In the Business Process Framework (eTOM)

The items concerned by the Status are:

- Domain: applies to everything inside the domain
- Process: applies to lower level processes of the specific process

In the Business Process Framework model the Status is specified as follows:

- Released: nothing specific
- Preliminary: with a "preliminary" notation on the domain or process
- Draft: with a "draft" notation on the domain or process
- Not Fully Developed: with a "not fully developed" notation on the domain or process
  (a field will be added to the model to hold the notation)

In the Business Process Framework Guidebooks the Status is specified as follows:

- Released: nothing specific
- Preliminary: with a "preliminary" notation on the domain or process
- Draft: Not published
- Not Fully Developed: with a "not fully developed"

In the Business Process Framework Map:

- Released: black font in a white rectangle
- Preliminary: black font in a medium-grey rectangle
- Draft: not presented
- Not Fully Developed: black font in a dark-grey rectangle

2.5.2 In the Information Framework (SID)

The items concerned by the Status are:

- Domain: applies to everything inside the Domain
- ABE: applies to everything inside the ABE
- BE: applies to all attributes that describe the BE

In the model, the Status is specified as the following:

- Released: nothing specific
- Preliminary: with a “preliminary” stereotype on the package and entities
- Draft: with a “draft” stereotype on the package and entities
- Not Fully Developed: with a “not fully developed” stereotype on the package and with at least a description

  All the generated models (browse model, HTML, XSD…) contain only the items with a maturity level Released, Preliminary and Not Fully developed

  The RSA model contains all items of all Status

In Guide Books:

- Released: nothing specific
- Preliminary: with “preliminary” included in the title or section
- Draft: does not appear in the guide book
- Not Fully Developed: Only appears in the Concepts and Principles document and are marked as “not fully developed”

In the Information Framework Map:

- Released: black font in a white rectangle
- Preliminary: black font in a medium-grey rectangle
- Draft: not presented
- Not Fully Developed: black font in a dark-grey rectangle

2.5.3 In the Application Framework (TAM)

The items concerned by the Status are:

- Domain: applies to everything to Domain

- Application: applies to all lower levels applications
- Functionality: applies to the functionality

In the Application Framework, in the model the status is specified as the following:

- Released: nothing specific
- Preliminary: with a “preliminary” notation on the domain or application or functionality
- Draft: with a “draft” notation on the domain or application or functionality
- Not Fully Developed: with a “not fully developed” notation on the domain or application or Functionality(a field will be added to the model to hold the notation)

In the Application Framework, in the Guidebooks, the Status is specified as the following:

- Released: nothing specific
- Preliminary: with a “preliminary” notation on the domain, application or functionality
- Draft: does not appear in the guide book
- Not Fully developed: with a “not fully developed” in the domain or application or functionality

In the Application Framework Map:

- Released: black font in a white rectangle
- Preliminary: black font in a medium-grey rectangle
- Draft: not presented
- Not Fully developed: black font in a dark-grey rectangle

From the 15.5 release onwards, the status of all Core Frameworks components will be reviewed to conform to the correct one.

3 Domain Definitions

3.1 Market/Sales Domain

The Market/Sales domain supports the sales and marketing activities needed to gain business from customers and potential customers. On the Sales side, this includes sales contacts/leads/prospects through to the sales-force and sales statistics. Market includes market strategy and plans, market segments, competitors and their products, through to campaign formulation. Market/Sales is framed in all Frameworx components (Business Process, Information and Application Framework) as the Market/Sales domain

3.2 Product Domain

The Product domain is concerned with the lifecycle of products offered to and procured by customers. This includes strategic portfolio plans, products offered, product performance, product usage statistics, as well as the product instances delivered to a customer.

3.3 Customer Domain

The Customer domain represents individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data. Customer also includes customer bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.

3.4 Service Domain

The Service Domain is concerned with the definition, development, and operational aspects of Services used to realize offerings to the market. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance and quality analysis. Finally, this domain also includes planning for future services, service enhancement or retirement and capacity.

3.5 Resource Domain

The Resource domain is concerned with the definition, development, and operational aspects of the applications, computing, and networks which represent the infrastructure of an enterprise. Resource management has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities to facilitate this association. The second is to ensure that Resources can support and deliver Products offered by the enterprise. Management of Resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of Resource Management is to enable strategy and planning processes to be defined.

3.6 Engaged Party Domain

The Engaged Party Domain includes all Engaged Party-oriented data and contract operations associated with an Engaged Party. Its scope encompasses, planning of strategies vs. Engaged Parties, handling of all types of contact with the Engaged Parties, the management of the relationship, and the administration of Engaged Parties data. The Engaged Party Domain also includes data and contract operations related to the Engaged Party Bills, disputes and inquiries.

3.7 Supplier/Partner Domain (only relevant for the Application Framework)

The Supplier/Partner domain encompasses planning of strategies for Supplier/Partners, handling of all types of contact with the Supplier/Partner, the management of the relationship, and the administration of Supplier/Partner data. It also supports bills, disputes and inquiries associated with a Supplier/Partner. While both Supplier and Partner are distinct third parties assisting with meeting an Enterprise's customer needs, they are different in the business model and risk sharing in the value chain under consideration. Partners have a greater risk share in an end customer transaction. Supplier/Partner is framed in the Application Framework by the Supplier/Partner Domain. With the necessary removal of the Supplier/Partner Domain in the Application Framework in the future, the current definition in this Core Frameworks Concepts and Principles document will be removed as well.

3.8 Enterprise Domain

The Enterprise domain provides support and sets policy for the overall business, enterprise or Service Provider. It also includes activities that are common to all enterprises across all industries such as accounting and human resource management.

3.9 Common Domain

The common domain is of different nature from the above defined ones as the processes, information data and applications described there do not necessarily have relationships. The elements in these common domains are related only to their respective framework.

In the Business Process Framework it is referred to as the Common process patterns (introduced in release 15.5). The Common Process Pattern domain contains processes, such as Cataloging, Capacity Management, and Configuration Specification and Configuration Management. They represent reusable components that can be used to define business processes, such as Product Capacity Management, Service Capacity Management, and Resource Capacity Management.

A key characteristic of Common Process Patterns domain is that they CANNOT execute on their own and ALWAYS require specialization (are used to define true business processes). For example, defining capacity has no meaning unless it is put in the context of a process, such as defining product capacity.

In the Information Framework it is referred to as the Common Business Entities domain defined as follows: The Common Business Entities Domain represents business entities shared across two or more other domains. As such, these business entities are not "owned" by any particular domain. In some cases a common business entity represents a generic abstraction of other real-world business entities. For example, Business Interaction is an abstraction (super-class) of business entities such as Customer Order and Supplier/Partner SLA.

In the Application Framework it is referred to as a cross domain, that includes specifically two common applications (Catalog Management and Fallout Management). The notion of common domain does not represent the broad dimension and purpose that it has in the Information Framework.

4 Information Framework to Business Process Framework Mappings – L1 ABEs to Core Processes

4.1 Principles for Mapping

The following principles and delimitations are used for coupling Information Framework ABEs to Business Process Framework business processes.

- When possible, an ABE is coupled to only one core process. This specific process is denoted as "primary".
- Primary core processes manage the complete life cycle of an ABE, by creating, reading, updating, and deleting (CRUD) entity instances contained within the ABE.
- Information is considered to flow both in the process layer as well as the systems (read ABE) layer
- If a one-to-one pairing is not attainable, an ABE is mapped to at most two core processes. If an ABE is mapped to two core processes, both core processes are considered to be primary.
- Processes requiring information feeds from ABEs or read an ABE's entities to function properly are termed as "secondary".
- In some cases a secondary process may update an ABE's entities. While this is permissible care must be taken. This is because it then begins to bind the secondary process more closely with the primary process, which means the two processes must be used together to manage the lifecycle of the ABE's entities. Often the secondary process can initiate a flow to the primary processes lower level process that updates an entity or entities rather than updating the entity or entities.

Regarding ABEs Supporting Two Core Processes.

Some ABEs are coupled to two or more core processes. This occurs as the result of:

- *Duality phenomenon*: implies that an arbitrary ABE supports two primary level 2 Business Process Framework processes, whereas the two primary processes belong to separate process areas. Thus, three combinations are available, [1] Operations & SIP, [2] Operations & EM or [3] SIP & EM. The duality phenomenon takes place because the ABE in question is regarded as unclear in terms of "core functionality", since it does not explicitly support only one primary level 2 Business Process Framework process.
- *Ambiguity phenomenon*: is similar to the duality phenomenon, however, here an arbitrary ABE supports two primary level 2 Business Process

Framework processes which both belong to only one of the three available process areas.

These phenomena may have any number of causes, such as:

- The ABE may be separated into two ABEs, each supporting a single level 2 Business Process Framework process.
- There may be a missing ABE that supports one of the level 2 Business Process Framework processes.
- There may be a missing Business Process Framework level 2 process that aggregates the two processes.
- A mistake could have been made in the mapping.

The phenomena will be investigated and, if possible, resolved in subsequent phases of the Information Framework project via the continued collaboration between the Information Framework team and the Business Process Framework team.

Business Process Framework Vertical Process Groupings

The column "eTOM Vertical" already present in former versions of documents GB922-CP and GB942MAP, has been preserved in this current version as an additional reference for the mapping of processes and ABEs. The following are the list of vertical process groupings used:

S&C - Strategy & Commit

ILM – Infrastructure Lifecycle Management

PLM – Product Lifecycle Management

OS&R – Operations Support & Readiness

F – Fulfilment

A – Assurance

B&RM – Billing & Revenue Management

FAB – Fulfilment, Assurance, Billing & Revenue Management (combination of the three verticals in Operations)

4.2 Market_Sales Domain

The Market/Sales Domain includes data and contract operations that support the sales and marketing activities needed to gain business from customers and potential customers. On the Sales side, this includes sales contacts/leads/prospects through to the sales-force and sales statistics. Market includes market strategy and plans, market segments, competitors and their products, through to campaign formulation.

Market_Sales Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Competitor ABE Identifies other providers who compete in the same market segments, accumulates intelligence about the competitors, including products (price, Key Performance Indicators and so forth).	S&C	Market Strategy & Policy	Enterprise Performance Assessment Product & Offer Portfolio Planning
Contact_Lead_Prospect ABE Provides the ability to track sales leads through their life cycle up until the time the leads become customers, including lead and contact information, sales prospects, proposals made to potential customers, and the amount of potential revenue the leads represent in the form of a sales pipeline.	F	Contact/Lead/Prospect Management	Manage Prospect Selling
Market Segment ABE Supports market segments, market statistics, and forecasts.	S&C	Market Strategy & Policy	Sales Development Product & Offer Portfolio Planning Product Specification & Offering Development & Retirement
Market Strategy & Plan ABE Supports the business plans and strategies on how to address the market with appropriate products and channels.	S&C	Market Strategy & Policy	Resource Strategy & Planning Product & Offer Portfolio Planning Party Strategy & Planning
Marketing Campaign ABE Supports marketing new or existing product offerings to identified target markets. For example, the launch of a pre-paid product with multiple promotions across distribution channels, market segments and so forth; a new campaign for an existing product; a re-launch of a campaign for an existing product.	PLM/OS&R	Marketing Campaign Management	Marketing Communications Marketing Fulfillment Response Market Research Sales Development
Sales Channel ABE Keeps track of distribution channels and sales activities, sales quotas, sales contests, commission/bonus plans, commissions/bonuses, and maintains groups of individuals that make up the sales force.	PLM	Sales Development	Sales Channel Management Contact/Lead/Prospect Management Market/Sales Support & Readiness
Sales Statistics ABE Maintains sales forecasts, new service requirements, customer needs, and customer education, as well as calculating key performance indicators about Sales & Marketing revenue and sales channel performance.	OS&R	Market Sales Support & Readiness	Market Strategy & Policy Sales Development Product Specification & Offering Development & Retirement

4.3 Product Domain

The Product Domain is concerned with the lifecycle of products and information and contract operations related to products' lifecycle. The Domain contains Aggregate Business Entities that deal with the strategic portfolio plans, products offered, product performance, product usage statistics, as well as the product instances delivered to a customer.

Product Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Loyalty ABE A loyalty program is one of the tools used by the loyalty process to retain customers. The Loyalty ABE contains all entities useful to specify and instantiate loyalty programs. A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply. Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not. A LoyaltyProgramProduct is a type of ProductComponent and described by a LoyaltyProgramProdSpec.	FAB	Enable Retention & Loyalty	Customer Information Management Customer Interaction Management Problem Handling Bill Inquiry Handling Customer QoS/SLA Management
Product ABE Represents an instance of a product offering subscribed to by a party, such as a customer, the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Product ABE also tracks the services and/or resources through which the product is realized.	F	Order Handling	Selling Bill Payments & Receivables Management Bill Inquiry Handling Charging Verify Product Configuration Problem Handling Customer QoS/SLA Management Bill Invoice Management SM&O Support & Readiness Develop Detailed Service Design Create Service Orders Service Configuration & Activation Service Problem Management Service Quality Management Service Guiding & Mediation Create Resource Orders
Product Configuration ABE The definition of how a Product operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s), ProductSpec(s), ServiceSpec(s) as well as a representation of how a Product operates or functions in terms of characteristics and related Resource(s), Product(s), Service(s).	F	Product Configuration Management	Problem Handling Product Performance Management Product Support & Readiness
Product Offering ABE Represents tangible and intangible goods and services made available for a certain price to the market in the form of product catalogs. This ABE is also responsible for targeting market segments based on the appropriate market strategy.	PLM	Product Specification & Offering Development & Retirement / Product Offering Development & Retirement	Marketing Communications Marketing Fulfillment Response Selling Order Handling Product & Offer Capability Delivery
Product Performance ABE The Product Performance ABE handles product performance goals, the results of end-to-end product performance assessments, and the comparison of assessments against goals. The results	A	Product Performance Management	Customer QoS/SLA Management Service Quality Management Resource Performance Management Product Specification & Offering Development & Retirement

The Product Domain is concerned with the lifecycle of products and information and contract operations related to products' lifecycle. The Domain contains Aggregate Business Entities that deal with the strategic portfolio plans, products offered, product performance, product usage statistics, as well as the product instances delivered to a customer.			
Product Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
may include the identification of potential capacity issues.			Enterprise Performance Assessment
Product Specification ABE Defines the functionality and characteristics of product offerings made available to the market.	PLM	Product Specification & Offering Development & Retirement / Product Offering Development & Retirement	SM&O Support & Readiness Develop Detailed Service Design Create Service Orders Service Configuration & Activation Create Resource Orders Product & Offer Capability Delivery
Product Test ABE A product test is a function performed on a product that results in measures being produced that reflect the functioning of the entity under test.	F	Product Configuration Management	Product Specification & Offering Development & Retirement Problem Handling
Product Usage ABE Represents usage of products associated with Customers used for charging that may appear on a Customer Bill.	OS&R	Customer Support & Readiness	Problem Handling Customer QoS/SLA Management Enterprise Performance Assessment Product Specification & Offering Development & Retirement
Strategic Product Portfolio Plan ABE Is concerned with the plans of the product port-folio, which product offerings to make available to each market segment and the plans to development and deploy product offerings, as well as retirement of products.	S&C	Product & Offer Portfolio Planning	Market Strategy & Policy Product Specification & Offering Development & Retirement

4.4 Customer Domain

The Customer Domain includes all data and contract operations associated with individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data. The Customer Domain also includes data and contract operations related to the customer bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.

Customer Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Applied Customer Billing Rate ABE Deals with the correlation of related usage for subsequent rating, rates applied to the usage (both regulated and non-regulated), discounts to usage, and any taxes due on the rated usage.	B&RM	Charging	Bill Payments & Receivables Management Bill Inquiry Handling Enable Retention & Loyalty Bill Invoice Management
Customer ABE Is the focus for the Customer domain? Customer data is the enterprise's knowledge of the customer and accounts held by a customer.	F	Selling	Marketing Communications Bill Payments & Receivables Management Customer Interaction Management Order Handling Enable Retention & Loyalty Problem Handling Customer QoS/SLA Management Bill Invoice Management
Customer Bill ABE Handles real-time and non-real-time Call Detail Records (CDRs) and other sources of data that result in invoice items. The Customer Bill ABE also represents the format of a bill, schedule the production of bills, customer invoicing profiles, all the financial calculations necessary to determine the total of the bill (except for rating and rating discounts), and credits and adjustments to bills.	B&RM	Bill Invoice Management	Enable Retention & Loyalty Bill Payments & Receivables Management Bill Inquiry Handling
Customer Bill Collection ABE Handles credit violations, actions for overdue debts, and facility billing audits.	B&RM	Bill Payments & Receivables Management	Enable Retention & Loyalty Bill Inquiry Handling
Customer Bill Inquiry ABE Represents invoice inquiries associated with invoices sent to customers and handles disputes and adjustments on individual charges, invoices, and accounts.	B&RM	Bill Inquiry Handling	Enable Retention & Loyalty Customer Management
Customer Interaction ABE Represents communications with customers, and the translation of customer requests and inquiries into appropriate "events" such as the creation of a customer order, the creation of a customer bill inquiry, or the creation of a customer problem.	FAB	Customer Interaction Management	Selling Bill Payments & Receivables Management Bill Inquiry Handling Order Handling Enable Retention & Loyalty Problem Handling Customer QoS/SLA Management Bill Invoice Management
Customer Order ABE Handles single customer orders and the various types thereof, such as regulated and non-regulated orders.	F	Order Handling	Selling Enable Retention & Loyalty Service Configuration & Activation Problem Handling Bill Inquiry Handling
Customer Problem ABE Focuses on technical assistance and problem handling for customers.	A	Problem Handling	Enable Retention & Loyalty Bill Inquiry Handling Service Configuration & Activation
Customer SLA ABE	A	Customer QoS/SLA Management	Selling

The Customer Domain includes all data and contract operations associated with individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data. The Customer Domain also includes data and contract operations related to the customer bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.			
Customer Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Is a special case of the Service Level Agreement ABE where an involved party in the agreement is a Customer. See the Agreement ABE in the Common Business Entity Domain for details.			Bill Payments & Receivables Management Bill Inquiry Handling Enable Retention & Loyalty
Customer Statistic ABE Represents the analysis of customer usage patterns, customer profitability statistics and churn and retention statistics.	OS&R FAB	Enable Retention & Loyalty Customer Support & Readiness	Selling Bill Payments & Receivables Management Bill Inquiry Handling Order Handling Problem Handling Customer QoS/SLA Management Bill Invoice Management

4.5 Service Domain

The Service Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of Services provided by an NGOSS system. Entities in this domain support various Business Process Framework processes that deal with the definition, development and management of services offered by an enterprise. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance, quality analysis, and rating. Finally, this domain also includes entities to perform planning for future offerings, service enhancement or retirement, and capacity.

Service Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Service ABE The Service ABE contains entities that are used to represent both customer-facing and resource-facing types of services. Entities in this ABE provide different views to examine, analyze, configure, monitor and repair Services of all types. Entities in this ABE are derived from Service Specification entities.	F	Service Configuration & Activation	Service Capability Delivery Service Development & Retirement SM&O Support & Readiness Service Problem Management Service Quality Management Service Guiding & Mediation
Service Configuration ABE The Service Configuration ABE contains entities that are used to represent and manage configurations of CustomerFacingService and ResourceFacingService entities. This set of entities also provides details on how the configuration of each of these types of Services can be changed. The entities in this ABE depend on entities in the Resource Domain, which provide the physical and logical infrastructure for implementing a Service. They all define dependencies between a higher-level Service and any sub-Services that are used by the higher-level Service.	F	Service Configuration & Activation	Service Capability Delivery SM&O Support & Readiness Resource Provisioning
Service Performance ABE The Service Performance ABE collects, correlates, consolidates, and validates various performance statistics and other operational characteristics of customer and resource facing service entities. It provides a set of entities that can monitor and report on performance. Each of these entities also conducts network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation. Entities in this ABE also manage the traffic generated by a Service, as well as traffic trend analysis. This is important for newer technologies that separate data, control and management functions for a given Service.	A	Service Quality Management	Service Development & Retirement SM&O Support & Readiness Service Problem Management Resource Performance Management
Service Problem ABE The Service Trouble ABE manages faults, alarms, and outages from a Service point-of-view. This is then correlated to trouble tickets, regardless of whether the cause is physical or logical. Other entities in this ABE are used to direct the recovery from each of these three types	A	Service Problem Management	Problem Handling SM&O Support & Readiness Service Quality Management Resource Trouble Management

The Service Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of Services provided by an NGOSS system. Entities in this domain support various Business Process Framework processes that deal with the definition, development and management of services offered by an enterprise. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance, quality analysis, and rating. Finally, this domain also includes entities to perform planning for future offerings, service enhancement or retirement, and capacity.			
Service Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
of problems. They provide the ability to associate Resource faults and alarms to degradation and outages of Services that run on those Resources. These functions are independent of the Resources and technologies used to build the Service. A third set of entities in this ABE is used to differentiate between customer-reported problems and network-induced problems.			
Service Specification ABE The Service Specification ABE contains entities that define the invariant characteristics and behavior of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the invariant characteristics and behavior defined in its associated template. Entities in this ABE focus on adherence to standards, distinguishing features of a Service, dependencies (both physical and logical, as well as on other services), quality, and cost. In general, entities in this ABE enable Services to be bound to Products and run using Resources.	PLM	Service Development & Retirement	Service Capability Delivery SM&O Support & Readiness Service Configuration & Activation Service Problem Management Service Quality Management Service Guiding & Mediation
Service Strategy & Plan ABE The Service Strategy and Plan ABE contains entities that are used to address the need for enhanced or new Services, as well as the retirement of existing Services, by the enterprise. These entities have a strong dependency to both entities in the Resource and Product domains. Resulting efforts, such as deciding what Resources to use to host a Service, or what Services are used to support new Product Specifications, are also supported, as are service demand forecasts.	S&C	Service Strategy & Planning	Service Capability Delivery Service Development & Retirement Resource Strategy & Planning Product & Offer Portfolio Planning
Service Test ABE The Service Test ABE contains entities that are used to test customer and resource facing service entities. These entities are usually invoked during installation, as a part of trouble diagnosis, or after trouble repair has been completed.	F	Service Configuration & Activation	Service Problem Management Service Quality Management
Service Usage ABE The Service Usage ABE collects Service consumption data, and generates Service usage records, for use by other business entities. The entities in this ABE provide physical, logical, and network usage information.	B&RM	Service Guiding & Mediation	Service Development & Retirement Service Quality Management Resource Data Collection & Distribution

4.6 Resource Domain

The Resource Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of an NGOSS system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure.
The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.

Resource Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Resource ABE The Resource ABE contains entities that are used to represent the various aspects of a Resource. This includes four sets of entities that represent: the physical and logical aspects of a Resource; show how to aggregate such resources into aggregate entities that have physical and logical characteristics and behavior; and show how to represent networks, sub-networks, network components, and other related aspects of a network.	F	Resource Provisioning	Resource Capability Delivery RM&O Support & Readiness Resource Data Collection & Distribution Resource Trouble Management Resource Performance Management
Resource Configuration ABE The Resource Configuration ABE contains entities that are used to represent and manage configurations of PhysicalResource, LogicalResource, and CompoundResource entities. It should be noted that configurations themselves are managed entities. This set of entities also provides details on how the configuration of each of these types of resources is changed in order to meet product, service, and resource requirements, including activation, deactivation, and testing. Areas covered include verifying resource availability, reservation and allocation of resource instances, configuring and activating physical and logical resource instances, testing and updating of the resource inventory database.	F	Resource Provisioning	Resource Capability Delivery RM&O Support & Readiness
Resource Performance ABE The Resource Performance ABE collects, correlates, consolidates, and validates various performance statistics and other operational characteristics of Resource entities. It provides a set of entities that can monitor and report on performance. The entities in this ABE provide physical, logical, and performance information. Each of these entities also conducts network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Resource degradation. Entities in this ABE also manage traffic in a	A	Resource Performance Management	Resource Development & Retirement RM&O Support & Readiness Resource Data Collection & Distribution

The Resource Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of an NGOSS system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure.
The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.

Resource Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Resource. This includes statistics defining Resource loading, and traffic trend analysis.			
Resource Specification ABE The Resource Specification ABE contains entities that define the invariant characteristics and behavior of each type of Resource entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the invariant characteristics and behavior defined in its associated template.	PLM	Resource Development & Retirement	Resource Capability Delivery RM&O Support & Readiness Resource Provisioning Resource Data Collection & Distribution Resource Trouble Management Resource Performance Management
Resource Strategy & Plan ABE The Resource Plan ABE is used to plan networks and resource elements both initially and for growth. It will coordinate both logical and physical resource growth. Inputs are budgets from business sources, service forecasts, current and projected network utilization, new technologies, and retiring technologies. It handles the lifecycle (installation, modification, removal, and retirement) for both logical and physical resources.	S&C	Resource Strategy & Planning	Service Strategy & Planning Resource Capability Delivery Resource Development & Retirement RM&O Support & Readiness
Resource Test ABE The Resource Test ABE contains entities that are used to test PhysicalResources, LogicalResources, CompoundResources, and Networks. These entities are usually invoked during installation, as a part of trouble diagnosis, or after trouble repair has been completed.	F	Resource Provisioning	Resource Capability Delivery Resource Trouble Management Resource Performance Management
Resource Topology ABE The Resource Topology ABE contains entities that define physical, logical, and network topological information. This information is critical for assessing the current state of the network, as well as providing information on how to fix problems, tune performance, and in general work with the network (both as a whole and with its components). Each of these topological views provides its own physical, logical, or network related information that can be used to manage one or more layers in a layered network.	ILM	Resource Capability Delivery	Resource Development & Retirement RM&O Support & Readiness

The Resource Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of an NGOSS system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure. The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.			
Resource Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Resource Trouble ABE The Resource Trouble ABE manages problems found in allocated resource instances, regardless of whether the problem is physical or logical. Entities in this ABE detect these problems, act to determine their root cause, resolve these problems and maintain a history of the activities involved in diagnosing and solving the problem. Detecting problems can be done via software (e.g. responding to an alarm) and/or by hardware (e.g. a measurement or probe) and/or manually (e.g. visual inspection). This includes tracking, reporting, assigning people to fix the problem, testing and verification, and overall administration of repair activities.	A	Resource Trouble Management	Service Problem Management RM&O Support & Readiness Resource Performance Management
Resource Usage ABE The Resource Usage ABE collects Resource consumption data, and generates Resource usage records, for use by other business entities. The entities in this ABE provide physical, logical, and network usage information.	FAB	Resource Data Collection & Distribution	Manage Billing Events Service Guiding & Mediation

4.7 Engaged Party Domain

The Engaged Party Domain includes all data and contract operations associated with individuals or organizations that are of interest to an enterprise. It represents all types of contact with the party, the management of the relationship, and the administration of party data. The Engaged Party Domain also includes data and contract operations related to the engaged party bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.

Engaged Party Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Additional Party Entities ABE Represents entities that are similar to those in the Customer ABE, such as PartyAccount and PartyAccountContact. This ABE was added to maintain backward compatibility and conformance of the Party ABE rather than include these entities within it.	FAB	Party Relationship Development & Retirement	Party Strategy & Planning Party Offering Development & Retirement Party Support & Readiness Party Order Handling Party Problem Handling Party Performance Management Party Revenue Management
Agreement ABE Represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other business entities, such as products, services, and resources and/or their specifications.	ILM, PLM & OS&R F	Party Agreement Management Selling	Party Interaction Management Party Performance Management Party Revenue Management
Party ABE The Party ABE represents the abstract concept of organization or individual that can play varying roles during interactions with an enterprise. Roles include customer, supplier/partner, employee, and so forth.	FAB	Party Relationship Development & Retirement	Party Relationship Management Party Demographic Collection Party Problem Handling Party Order Handling Party Performance Management Party Interaction Management Party Bill Inquiry Handling Party Revenue Management
Party Interaction ABE Represents communications with parties, and the translation of requests and inquiries into appropriate "events" such as the creation of an order, the creation of a bill inquiry, or the creation of a problem.	FAB	Party Interaction Management	Party Demographic Collection Party Problem Handling Party Order Handling Party Performance Management Party Bill Inquiry Handling Party Revenue Management
Party Order ABE Handles party orders and the various types thereof, such as regulated and non-regulated orders.	F	Party Order Handling	Party Problem Handling Party Bill Inquiry Handling
Party Privacy ABE The Party Privacy Profile ABE contains all entities used by the Party Privacy Management process for specifying the information concerned by Privacy rules, the Privacy rules themselves, and the choices made by Parties for their own Privacy.	OS&R	Party Privacy Management	Party Interaction Management Party Agreement Management
Party Problem ABE Focuses on technical assistance and problem handling reported to and by other parties.	A	Party Problem Handling	Party Order Handling Party Bill Inquiry Handling Party Interaction Management Party Relationship Development & Retirement
Party Product Specification & Offering ABE The Party Product Specification and Offering ABE represents the involvement	PLM	Party Offering Development & Retirement	Party Tender Management Party Agreement Management

The Engaged Party Domain includes all data and contract operations associated with individuals or organizations that are of interest to an enterprise. It represents all types of contact with the party, the management of the relationship, and the administration of party data. The Engaged Party Domain also includes data and contract operations related to the engaged party bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.			
Engaged Party Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
parties playing roles have with ProductSpecifications ProductOfferings, such as ordered from and billed by.			
Party Revenue ABE The Party Revenue ABE represents enterprise revenue in the form of applied billing rates, bills, payments, and settlements.	B&RM	Party Revenue Management	Enterprise Risk Management Financial & Asset Management Party Agreement Management
Party SLA ABE A Service Level Agreement ABE with one or more other parties. See the Agreement ABE in the in the Engaged Party domain for additional details.	A	Party Performance Management	Resource Performance Management Service Quality Management Customer QoS/SLA Management Party Support & Readiness
Party Statistic ABE Represents the analysis of party usage patterns, profitability statistics.	OS&R/FAB	Party Support & Readiness Party Interaction Management	Party Strategy & Planning Party Performance Management
Party Strategy ABE The strategies and the planning of the business relation with the other parties with input from other ABEs, such as MarketSales, Party Performance and Competitor Analysis.	S&C	Party Strategy & Planning	Party Offering Development & Retirement Party Tender Management

4.8 Enterprise Domain

The Enterprise Domain provides support and sets policy for the overall business, enterprise or Service Provider. It also includes activities that are common to all enterprises across all industries such as accounting and human resource management.			
Enterprise Domain ABEs	eTOM Vertical	Primary Process	Secondary Process
Enterprise Effectiveness ABE	NA	Enterprise Effectiveness Management	Party Performance Management Customer Interaction Management Customer Support & Readiness
Enterprise Risk ABE	NA	Enterprise Risk Management	Party Relationship Development & Retirement Party Performance Management
Workforce ABE	OS&R/EM	Workforce Management / Human Resources Management	Party Performance Management Enterprise Effectiveness Management

4.9 Common Business Entities Domain

The Common Business Entities Domain represents business entities shared across two or more other domains. As such, these business entities are not "owned" by any particular domain. In some cases a common business entity represents a generic abstraction of other real-world business entities. For example, Business Interaction is an abstraction (super-class) of business entities such as Customer Order and Supplier/Partner SLA.			
Common Business Entities Domain ABEs	eTOM Vertical	Primary Core Processes	Secondary Core Processes
Base Types ABE A small simple object, like money or a date range, whose equality isn't based on identity.	NA	NA	NA
Business Interaction ABE The Business Interaction ABE represents an arrangement, contract, or communication between an enterprise and one or more other entities such as individuals and organizations (or parts of organizations). Interactions take on the form of requests, responses, and notifications.	NA	NA	NA
Calendar ABE The Calendar ABE represents Entities used to provide time related functions. This includes scheduling, time conflicts and time based presentation support.	NA	NA	NA
Capacity ABE Capacity represents the ability to provide a measured capability of the network.	NA	Capacity Management	NA
Catalog ABE A catalog can be defined as a collation of items and arranging them in a particular manner based on the need. For example, a product catalog.	NA	Cataloging	NA
Configuration & Profiling ABE A Configuration (also referred to as a Profile) defines how a Resource, Service, or Product operates or functions. A Configuration may contain one or more parts (which is realized by using the Atomic/Composite pattern, but it is represented as a single entity – ConfigurationRelationship), and each part may contain zero or more fields. Each field may have attributes that are statically or dynamically defined. Some of these fields have fixed values, while others provide values from which a choice or choices can be made (e.g., using the EntitySpec/Entity and/or CharacteristicSpec/CharacteristicValue patterns).	NA	NA	NA
Event ABE The Event ABE contains entities that are used to represent events, their occurrence and their recording within systems.	NA	NA	NA
Federated Catalog ABE	NA	Cataloging	

The Common Business Entities Domain represents business entities shared across two or more other domains. As such, these business entities are not "owned" by any particular domain. In some cases a common business entity represents a generic abstraction of other real-world business entities. For example, Business Interaction is an abstraction (super-class) of business entities such as Customer Order and Supplier/Partner SLA.			
Common Business Entities Domain ABEs	eTOM Vertical	Primary Core Processes	Secondary Core Processes
The Federated Catalog ABE contains entities that are used, in addition to the Catalog Entity in the Catalog ABE, to model dynamic, heterogeneous and generic catalogs of entities.			
Location ABE The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations.	NA	NA	NA
Metric ABE The Metric ABE defines a pattern to standardize the description of Metrics. It's in vision that other projects use and specialize the Metric ABE in some way (creating sub-classes and adding new entities) such as PM (Performance Management), RA (Revenue Assurance), SLA (Service Level Agreement), etc.	EM	Enterprise Effectiveness Management / Enterprise Performance Assessment / Metric Management	NA
Performance ABE Represents a measure of the manner in which a Product and/or Service and/or Resource is functioning.	NA	NA	NA
Policy ABE Policy consists of a set of layered ABEs that define specifications (for example, templates) and definitions of Policy entities that can be used in managing the behavior and definition of entities in other Domains. Policy takes three primary forms. The first is the definition of how policy is used to manage the definition, change, and configuration of other entities. The second is the definition of how policy itself is managed. The third is how applications use policies to manage entities.	NA	NA	NA
Project ABE The Project ABE represents the tools used by project managers to ensure that enterprise objectives of quality, cost, and time are achieved by planning and scheduling work. It uses common industry definitions of Project, Work Breakdown Structure and Activity to provide support to project managers.	NA	NA	NA
Root ABE A set of common business entities that collectively serve as the foundation of the business view. This set of entities enables	NA	NA	NA

The Common Business Entities Domain represents business entities shared across two or more other domains. As such, these business entities are not "owned" by any particular domain. In some cases a common business entity represents a generic abstraction of other real-world business entities. For example, Business Interaction is an abstraction (super-class) of business entities such as Customer Order and Supplier/Partner SLA.			
Common Business Entities Domain ABEs	eTOM Vertical	Primary Core Processes	Secondary Core Processes
the entities in different domains of the SID Framework to be associated with each other, providing greater overall coherence to the information framework.			
Test ABE A test is a function performed on a product, service, or resource that results in measures being produced that reflect the functioning of the entity under test.	NA	NA	NA
Topology ABE The Topology ABE contains entities that are used to represent topological concepts that can be used to model a large variety of topological relations between entities ranging from dependencies to connectivity.	NA	NA	NA
Trouble or Problem ABE A description of a problem that can be shared between the service provider and the customer. Trouble or Problem is an indication that an entity (such as Resource, Service or Product) is no longer functioning according to the expected SLA.	NA	NA	NA
Trouble Ticket ABE Represents a record used for reporting and managing the resolution of Trouble or Problem.	NA	NA	NA
Usage ABE An occurrence of employing a Product, Service, or Resource for its intended purpose, which is of interest to the business and can have charges applied to it. It is comprised of characteristics, which represent attributes of usage.	NA	NA	NA
Users and Roles ABE User and Roles represents types of users and roles and their involvement in the usage of products, services, and resources.	NA	NA	NA

PART II – BUSINESS PROCESS FRAMEWORK

5 Introduction

The Business Process Framework is intended to deliver reusable process elements to enable the construction of best practice or implementation specific process flows to a wide variety of purposes.

5.1 What is the Business Process Framework?

The Business Process Framework is a reference framework or model for categorizing all the business activities that a service provider will use. It is NOT a service provider business model. In other words, it does not address the strategic issues or questions of who a service provider's target customers should be, what market segments the service provider should serve, what are a service provider's vision, mission, etc. A business process framework is one part of the strategic business model and plan for a service provider.

The Business Process Framework is better regarded as a framework, rather than a business process model, since its aim is to categorize the process elements and business activities so that these can then be combined in many different ways, to implement end-to-end business processes (e.g. fulfillment, assurance, billing) which deliver value for the customer and the service provider. Key concepts that form the basis for the Business Process Framework are outlined in Annex A, Business Process Framework Concepts, and readers that are not familiar with the Business Process Framework may wish to gain an initial view of these concepts, to provide context before reading the main document.

Previous Business Process Framework (eTOM) Releases have established the Business Process Framework as TM Forum member-approved, with global agreement from the highest conceptual level downwards, and has gone on to take account of real-world experience in applying the Business Process Framework, and to incorporate new detail in process decompositions, flows and business-to-business interaction. Beyond this, the Business Process Framework work has potential to develop further, in areas such as further lower-level process decompositions and flows, applications in specific areas of business, guidelines and assistance in using the Business Process Framework, cost and performance issues associated with the processes, etc. In addition, ongoing feedback from the industry, together with its linkage with the wider Frameworx program, can be used to guide future priorities for continuing work. It should be noted that the development of a full process framework is a significant undertaking, and the work must be phased over time based on member process priorities and member resource availability. This effect is visible in Business Process Framework's history, from the early work on a business process map that carried through to the Business Process Framework itself, broadening along the way to a total enterprise framework and the current Release.

A great many service providers, as well as system integrators, ASPs and other types of vendors, are working already with the Business Process Framework. They need an industry standard framework for procuring software and equipment, as well as to interface with other service providers in an increasingly complex network of business relationships. Many service providers have contributed their own process models because they recognize the need to have a broader industry framework that doesn't just address operations or traditional business processes.

The Business Process Framework is also used as a consensus tool for discussion and agreement among service providers and network operators. This encouraged convergence and general support for a broad common base in this area, which has been built on and extended with the Business Process Framework, to enable:

- Focused work to be carried out in TM Forum teams to define detailed business requirements, information agreements, business application contracts and shared data model specifications (exchanges between applications or systems) and to review these outputs for consistency
- Relating business needs to available or required standards
- A common process view for equipment suppliers, applications builders and integrators to build management systems by combining third party and in-house developments.

The anticipated result is that the products purchased by service providers and network operators for business and operational management of their networks, information technologies and services will integrate better into their environment, enabling the cost benefits of end-to-end automation. Furthermore, a common industry view on processes and information facilitates operator-to-operator operator-to-customer and operator-to-supplier/partner process interconnection, which is essential for rapid service provisioning and problem handling in a competitive global environment. This process interconnection is the key to digital services supply chain management in particular.

The Business Process Framework work also provides the definition of common terms concerning enterprise processes, sub-processes and the activities performed within each. Common terminology makes it easier for service providers.

Annex A, Terminology and Acronym Glossary, contains definitions of Business to negotiate with customers, third party suppliers, and other service providers.

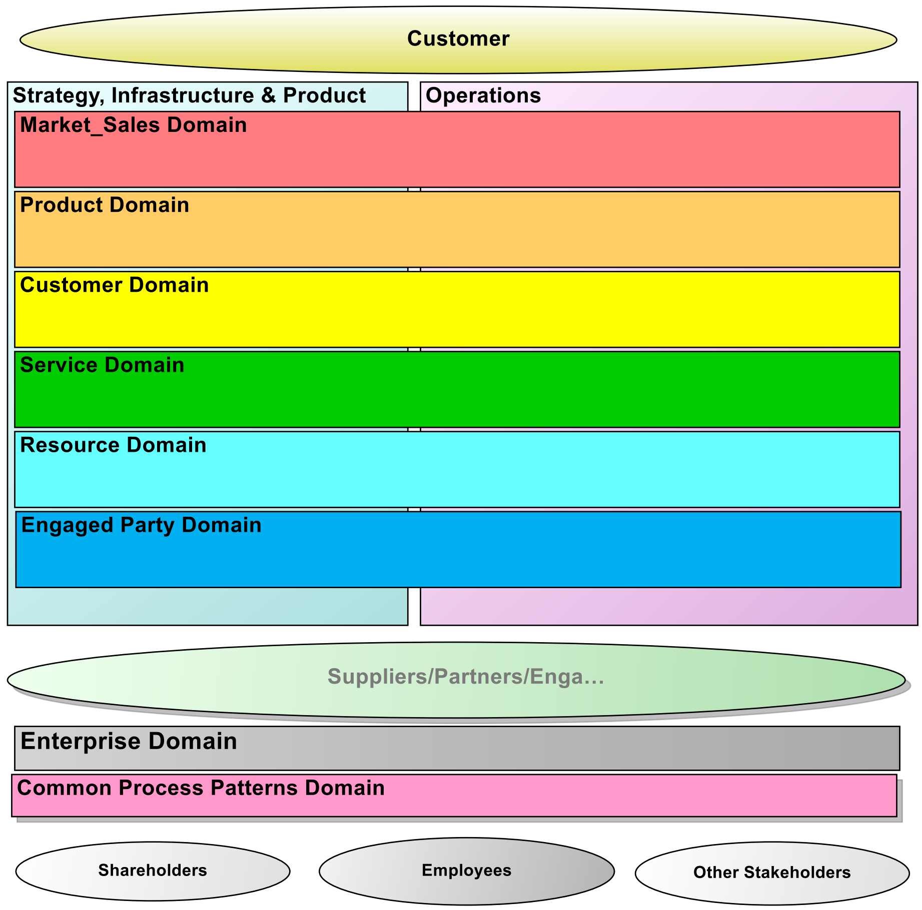

Figure 5-1 - Business Process Framework – Conceptual Model

Figure **5-1** displays the highest conceptual view of the Business Process Framework. This view provides an overall context that differentiates strategy and lifecycle processes from the operations processes in two large process areas or categories, seen as the two major boxes in the upper part of the diagram. It also differentiates the key management areas as horizontal layers or domains across these process areas. The third major process area, concerned with management of the enterprise itself, is shown as a separate domain in the lower part of the diagram. In addition, Figure **5-1** also shows the internal and external entities that interact with the enterprise (as ovals).

Figure 4-2 shows seven vertical process groupings that categorize the processes that are required to support customers and to manage the business. Amongst these Process Groupings, the focal point of the Business Process Framework is on the core customer operations processes of Fulfillment, Assurance and Billing (FAB). Operations Support & Readiness (OSR) is differentiated from FAB as typically these processes apply to multiple instances of product, service or resource at once. Outside of the Operations process area - in the Strategy, Infrastructure & Product (SIP) process area - the Strategy & Commit group, as well as the two Lifecycle Management groups, are differentiated. These are distinct because, unlike Operations, they do not directly support the customer, are intrinsically different from the Operations processes and work on different business time cycles.

The horizontal functional process groupings that depicts actually domains distinguish functional operations processes and other types of business functional processes, e.g., Marketing versus Selling, Service Development versus Service Configuration, etc. Amongst processes defined in these Horizontal domains, those on the left (that cross the Strategy & Commit, Infrastructure Lifecycle Management and Product Lifecycle Management vertical process groupings) enable, support and direct the work in the Operations process area. These processes form the apex of the process decomposition hierarchy we use to contain, find and define our processes.

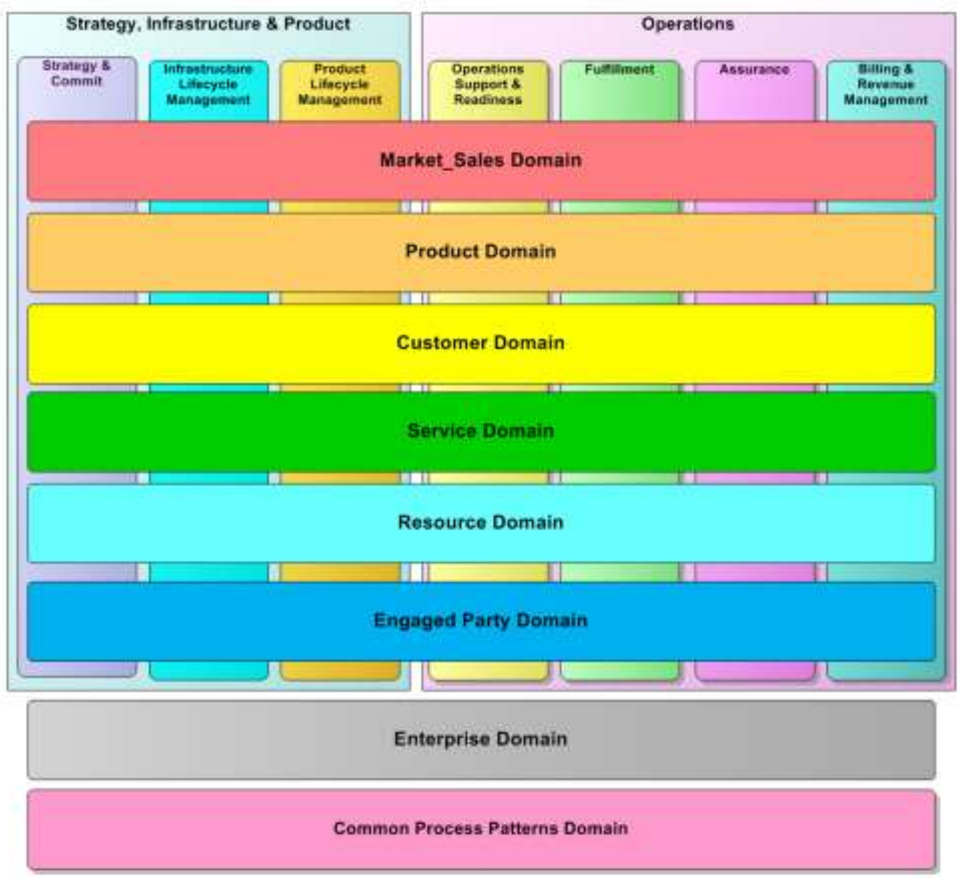

Figure 5-2 - Business Process Framework - domains and process groupings

By reference to Figure **5-2**, the Business Process Framework provides the following benefits:

- It develops a scope addressing all enterprise processes.
- It distinctly identifies marketing processes to reflect their heightened importance in a digital services world.
- It distinctly identifies Enterprise Management processes, so that everyone in the enterprise is able to identify their critical processes, thereby enabling process framework acceptance across the enterprise.
- It brings Fulfillment, Assurance and Billing (FAB) onto the high-level framework view to emphasize the customer priority processes as the focus of the enterprise.
- It defines an Operations Support & Readiness grouping, which relates to all the Operations functional process groupings. In integrating digital services and making customer self-management a reality, the enterprise has to understand the processes it needs to enable for direct and (more and more) online customer operations support and customer self-management.
- It recognizes three process groupings or categories within the enterprise that are distinctly different from operations processes by identifying the SIP processes category, i.e., Strategy & Commit, Infrastructure Lifecycle Management and Product Lifecycle Management.
- It recognizes the different cycle times of the strategy and lifecycle management processes and the need to separate these processes from the customer priority operations processes where automation is most critical. This is done by decoupling the Strategy & Commit and the two Lifecycle Management processes from the day-to-day, minute-to-minute cycle times of the customer operations processes.
- It moves from the older customer care or service orientation to a customer centered management orientation that emphasizes customer self-management and control, improving customer experience and increasing the value customers contributes to the enterprise and the use of information to customize and personalize to the individual customer. It adds more elements to this customer operations functional process grouping to represent better the selling processes
- It acknowledges the need to manage resources across technologies, (i.e., application, computing and network), by integrating the Network and Systems Management functional process into the Resource Domain. It also moves the management of IT into this functional process grouping as opposed to having a separate process grouping.
- It recognizes that the enterprise interacts with external parties, and that the enterprise may need to interact with process flows defined by external parties, as in digital services interactions.

5.2 Purpose of the Business Process Framework

The purpose of the Business Process Framework is to provide a standard process structure, terminology, and classification scheme for business processes within a Service Provider type of enterprise. The intent is to offer a foundation in the form of an enterprise-wide discipline for the development of business processes that enable consistent and reusable end to end process flows to be created, which in turn enable key information about any process to be captured and made available. It provides the basis for determining how process ownership can be assigned and the visibility of processes across the wider organization.

5.3 Scope of the Business Process Framework

The scope of the Business Process Framework is to focus on the business processes of a Service Provider enterprise. While having undeniable Communications industry roots, the Framework, like its cousin, ITIL–which initially arose in Military Logistics–has transcended its origins and is now deployed to address architecture concerns in many Digital or more general Service Providers.

6 Business Process Framework Concepts

The main purpose of this Chapter is to introduce a formal description of the Business Process Framework.

It should be noted that this framework was originally developed from the perspective of the single enterprise, but recognized that internal processes extend across the enterprise boundary to allow for interactions with external parties (customers, suppliers/partners and other parties).

In some cases these external interactions can be defined and controlled by the enterprise, and the existing Business Process Framework has assumed that the currently identified process elements would form part of the end-end inter-enterprise or enterprise-to-customer process interaction in these cases.

Several industries have developed inter-enterprise business process frameworks which specify the structure and flow of process interactions between multiple enterprises. PaDIOM (Partner, Design, Integrate, Operate and Monetize) is the TM Forum Lifecycle method for linking business and technical collaboration activities to ensure reuse of concepts, specifications and reusable service components across many partnerships. These mechanisms lead to agile, repeatable, reusable industrial scale B2B2X solutions. For more information on this topic, please refer to B2B2X on the TM Forum website.

6.1 Introduction

The Business Process Framework considers the service provider's (sometimes termed Communications Service Provider, or CSP) enterprise, and positions this within its overall business context: i.e. the business interactions and relationships, which allow the CSP to carry on its business with other organizations.

This section introduces the Business Process Framework and explains its structure and the significance of each of the process areas within it. It also shows how the Business Process Framework structure is decomposed to lower-level process elements. This explanation is useful for those who decide where and how an enterprise will use the Business Process Framework, and those who may be modifying it for use in their enterprise.

To assist the reader in locating the process area concerned within the Business Process Framework, a graphical icon of the Business Process Framework structure, alongside the text, is provided to draw attention to the relevant framework area. This is highlighted in red to indicate the focus of the associated text or discussion.

6.1.1 Business Process Framework Conceptual View

The Conceptual Structure view provides an overall context that differentiates strategy and lifecycle processes from operations processes in two large process areas, seen as the two large boxes towards the top of the diagram, together with a third area beneath which is concerned with enterprise management. It also identifies the key functional process structures as horizontal domains across the two upper process areas.

6.1.2 Business Process Framework CxO Level View (Level 1)

Below the conceptual level, the Business Process Framework (eTOM) is decomposed into a set of domains that regroups process, which provide a first level of detail at which the entire enterprise can be viewed (see Figure **6-1**). These process groupings or domains are considered from the perspective of the CEO, CIO, CTO, etc., in that the performance of these processes determines the success of the enterprise.

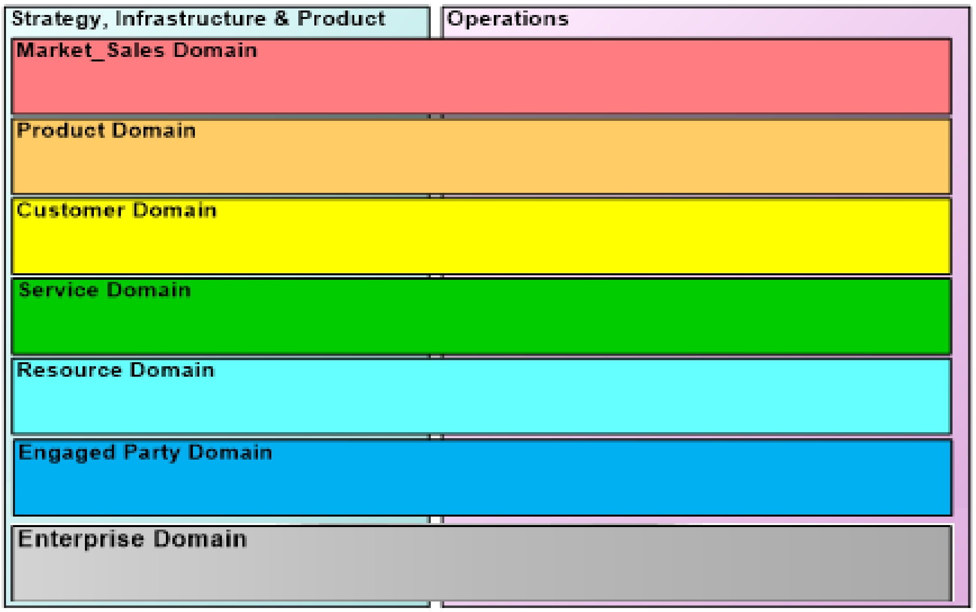

Figure 6-1 - Business Process Framework Domain View

The Business Process Framework is defined as generically as possible, so that it is independent of organization, technology and service. The eTOM is basically intuitive, business driven and customer focused. To reflect the way businesses look at their processes, the eTOM supports two different perspectives on the grouping of the detailed process elements:

- ***Domains,*** which represent a view of functionally-related processes within the business, such as those involved in managing contact with the customer or in managing the supply chain. This structuring by domains is useful to those who are responsible for creating the capability that enables, supports or automates the processes. The horizontal functional process groupings can therefore often represent the CIO's view of the Business Process Framework. The IT teams will look at groups of IT functions which tend to be implemented together e.g. the front-of-house applications in the Customer domain process grouping, back-of-house applications in the Service domain which focus on managing information about the services that are packaged for sale to customers, or the network management applications in the Resource domain which focus on the technology which delivers the services. Typical organization workgroups also tend to align with domains as the required knowledge and skills tend to be contained into these functional processes, e.g. the front-of-house workgroups in the Customer domain, back-of-house workgroups customers in the Service domain which focus on managing information about the services that are packaged for sale to customers, or the network management workgroups services in the Resource domain which focus on the technology which delivers the services.

- ***Vertical Process groupings,*** which represent a view of related processes within the business, such as those involved in the overall billing flows to customers. This view is important to those people who are responsible for changing, operating and managing the processes. These processes tend to span organization boundaries, and so the effectiveness of these processes is an area of concern to senior management and particularly the CEO. The vertical groupings can therefore often represent the CEO's view of the Business Process Framework. These people are more interested in the outcomes of the process and how they effectively support customer needs in total - rather than worrying about the IT or the specific workgroups that need to work together to deliver the result. Particular end to end process flows have been found to transcend the boundaries of these vertical groupings, leading to a relaxation of what was previously intended by this structure to become a supplementary grouping to assist users with finding detail, rather than expecting a business significance to be depicted. However these grouping are useful to picture the process frameworks and have a feel for where level 2 processes belong.

The Business Process Framework was developed to help build and implement the processes for a service provider. It has been developed as a structured catalogue or hierarchical taxonomy of process elements which can be viewed in more and more detail. Since in any taxonomy each element must be unique, it was decided from the start that the primary top-level hierarchy of process elements would be the functional (horizontal) processes. The vertical groupings are arranged as an overlay on the horizontal providing a matrix view of the process framework.

When viewed in terms of the **Cross-Functional process groupings**, the Business Process Framework follows a strict hierarchy where every element is only associated with or parented to a single element at the next higher hierarchical level. In a taxonomy, any element must be unique, i.e. it must be listed only once. Figure 6-1 shows the horizontal functional process groupings into which the Business Process Framework is decomposed.

Additionally, the Business Process Framework is intended to help service providers manage their business process flows. With this in mind, the Business Process Framework shows how process elements have a strong association with one (or several) vertical business groups (e.g. Fulfillment, Assurance, Billing & Revenue Management, Product Lifecycle Management etc., which are introduced later in this Chapter). These **process groupings** are essentially overlays onto the hierarchical top-level horizontal groupings, because in a hierarchical taxonomy an element cannot be associated with or parented to more than one element at the next higher level. Note that the Business Process Framework decomposition hierarchy operates exclusively through the Horizontal domain described above, and that these Vertical process groupings therefore do not form part of the actual decomposition hierarchy. The Vertical process groupings should thus be viewed as ancillary views or arrangements of process elements, provided for information and navigation only, as "overlays" on the actual process hierarchy.

The overlay of the horizontal domains and the vertical process groupings forms the inherent matrix structure of the Business Process Framework. This matrix structure is the core of one of the innovations and fundamental benefits of the Business Process Framework. It offers for the first time a standard language and structure for the process elements that can be understood and used by both the people specifying and operating the end-to-end business, and also those people who are responsible for creating the capability that enables the processes (whether automated by IT or implemented manually by workgroups).

The integration of all these processes provides the enterprise-level process framework for the service provider. As process decomposition proceeds, each level is decomposed into a set of constituent process elements at the level below.

Thus, the Enterprise conceptual view resolves into seven vertical process groupings, as well as twelve horizontal functional process groupings represented by six layers. These vertical and horizontal process groupings represent alternative views relevant to different concerns about the way that processes should be associated. Note that we will see that these alternatives have been selected to yield a single, common view of the processes defined at the next level of decomposition, and hence do not represent a divergence in the modeling.

6.1.3 Operations Process Area

To be useful to a service provider, the Business Process Framework must help the service provider to develop and operate their business processes. This section shows how the matrix structure of the Business Process Framework offers for the first time a standard language and structure for the process elements that are understood and used by both the people specifying and operating the end-to-end business, and also those people who are responsible for creating the capability that enables the processes (whether automated by IT or implemented manually by workgroups).

6.1.3.1 Operations Category

The Operations (OPS) process area contains the direct operations groupings of Fulfillment, Assurance & Billing (the FAB process groupings), together with the Operations Support & Readiness grouping (see Figure 6-2). The FAB Category groupings are sometimes referred to as Customer Operations processes.

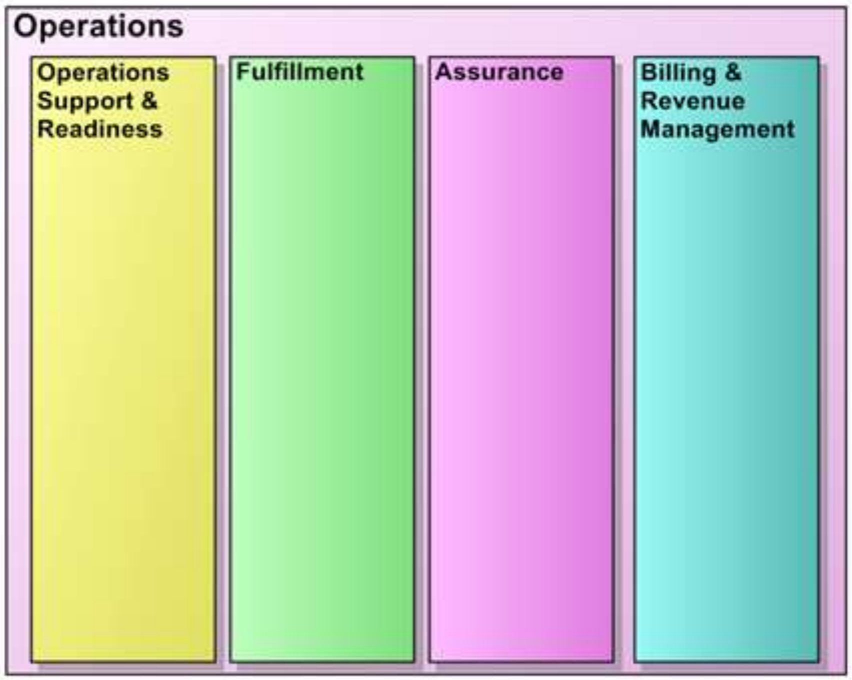

Figure 6-2 - Business Process Framework OPS Vertical Categories

Fulfillment: this grouping is responsible for providing customers with their requested products in a timely and correct manner. It translates the customer's business or personal need into a solution, which can be delivered using the specific products in the enterprise's portfolio. This process informs the customers of the status of their purchase order, ensures completion on time, as well as ensuring a delighted customer.

Assurance: this grouping is responsible for the execution of proactive and reactive maintenance activities to ensure that services provided to customers are continuously available and performing to SLA or QoS performance levels. It performs continuous resource status and performance monitoring to proactively detect possible failures. It collects performance data and analyzes them to identify potential problems and resolve them without impact to the customer. This process manages the SLAs and reports service performance to the customer. It receives trouble reports from the customer, informs the customer of the trouble status, and ensures restoration and repair, as well as ensuring a delighted customer.

Billing & Revenue Management: this grouping is responsible for the collection of appropriate usage records, determining charging and billing information, production of timely and accurate bills, for providing pre-bill use information and billing to customers, for processing their payments, and performing payment collections. In addition, it handles customer inquiries about bills, provides billing inquiry status and is responsible for resolving billing problems to the customer's satisfaction in a timely manner. This process grouping also supports prepayment for services.

For a high-level view of how the Business Process Framework can be used to create Fulfillment, Assurance and Billing & Revenue Management process flows, see GB921F.

In addition to these FAB process groupings, the OPS process area of the Business Process Framework contains a fourth vertical grouping: Operations Support & Readiness (see Figure 6-2).

Operations Support & Readiness: this grouping is responsible for providing management, logistics and administrative support to the FAB process groupings, and for ensuring operational readiness in the fulfillment, assurance and billing areas. In general, end-end processes in this grouping are concerned with activities that are less single customer specific than those in FAB, and which are typically concerned less with individual customers and services and more with ensuring the FAB groupings run effectively. A clear example of this type of processes is the staffing capacity management processes which are used to ensure efficient operation of call centers. They reflect a need in some enterprises to divide their processes between the immediate customer facing and real-time operations of FAB and other Operations processes which act as a “second-line” or “operations management back-room”. Not all enterprises will choose to employ this split, or to position the division in exactly the same place, so it is recognized that in applying the Business Process Framework in particular scenarios, the processes in Operations Support & Readiness and in FAB may be merged for day-to-day operation. Nevertheless, it is felt important to acknowledge this separation to reflect a real-world division that is present or emerging in many enterprises. The separation, definition and execution of the Operations Support & Readiness processes can be critical in taking advantage of digital services opportunities, and is particularly important for successful implementation of Customer Self-Management.

6.1.4 Strategy Infrastructure & Product Process Area

6.1.4.1 SIP Category

The Strategy and Commit, Infrastructure Lifecycle Management and Product Lifecycle Management process groupings, are shown as three vertical process groupings (see Figure 6-3). The Strategy and Commit process grouping provides the focus within the enterprise for generating specific business strategy and gaining buy-in within the business for this. Product Lifecycle Management process grouping drives and supports the provision of products to customers, while the Infrastructure Lifecycle Management process grouping delivers of new or enhanced infrastructure on which the products are based. Their focus is on meeting customer expectations whether as product offerings, the infrastructure that supports the operations functions and products, or the suppliers and partners involved in the enterprise’s offering to customers.

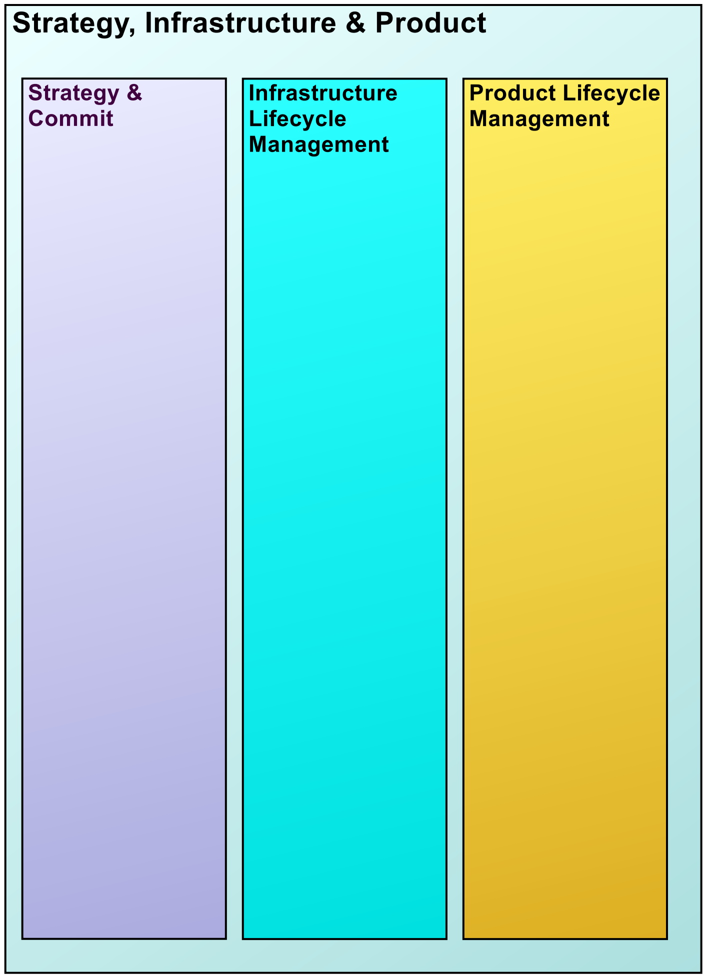

Figure 6-3 - Business Process Framework SIP Category Groupings

Strategy & Commit: this process grouping is responsible for the generation of strategies in support of the Infrastructure and Product Lifecycle processes. It is also responsible for establishing business commitment within the enterprise to support these strategies. This embraces all levels of operation from market, customer and products, through the services and the resources on which these depend, to the involvement of suppliers and partners in meeting these needs. Strategy & Commit processes are heavily focused on analysis and commitment management. These processes provide the focus within the enterprise for generating specific business strategy and gaining buy-in within the business to implement this strategy. Strategy & Commit processes also track the success and effectiveness of the strategies and make adjustments as required.

Lifecycle Management process groupings drive and enable core operations and customer processes to meet market demand and customer expectations. Performance of Lifecycle processes are viewed at the highest levels of the enterprise, due to their impact on customer retention and competitiveness. There are two end-to-end Lifecycle

Management processes introduced in the Business Process Framework, i.e., Infrastructure and Product. Both end-end processes have a development and deployment nature, in terms of introducing new infrastructure, or a new product. Infrastructure Lifecycle Management deals with development and deployment of new infrastructure, assessing performance of the infrastructure and taking action to meet performance commitments. Product Lifecycle Management deals with introducing new products, in the form of services delivered to customers, and assessing and taking action on product performance.

The Business Process Framework consciously decouples the Lifecycle Management processes from day-to-day operations processes represented by the Operations Processes (Operations Support & Readiness, Fulfillment, Assurance and Billing). In the past, some of these processes were not distinguished from the core operations framework and this sometimes resulted in some confusion and lack of guidance for designing processes. Lifecycle Management processes have different business cycles, different types of objectives for the enterprise and are inherently different processes than operations processes, i.e., enabling processes rather than operations processes. Mixing these processes with the customer operational processes diminishes focus on the Lifecycle Management processes. In addition, Lifecycle Management processes need to be designed to meet cycle time and other performance characteristics critical to the success of the enterprise, e.g., new product time to market, and infrastructure unit cost. The Lifecycle Management end-end processes interact with each other. The Product Lifecycle Management process group drives the majority of the direction for the Infrastructure Lifecycle Management process groups either directly or indirectly, for example. However Infrastructure Lifecycle Management vertical end-end processes are also driven by decisions within the Strategy & Commit process groups to deploy new infrastructure in support of new business directions. These processes prepare the customer and functional operations processes to support customer interaction for products, providing the infrastructure for the products to use and providing the supplier and partner interface structure for the enterprise offers. To enable and support customer and functional operations, these processes often have to synchronize for on-time and quality delivery.

Infrastructure Lifecycle Management: this process grouping is responsible for the definition, planning and implementation of all necessary infrastructures (application, computing and network), as well as all other support infrastructures and business capabilities (operations centers, architectures, etc.). This applies in connection with the resource layer or any other functional layer, e.g., CRM Voice Response Units, required to provide Information and Communications products to the Customer and to support the business. These process groups identify new requirements, new capabilities and design and develop new or enhanced infrastructure to support products. Infrastructure Lifecycle Management processes respond to needs of the Product Lifecycle Management processes whether unit cost reductions, product quality improvements, new products, etc.

Product Lifecycle Management: this process grouping is responsible for the definition, planning, design and implementation of all products in the enterprise's portfolio. The Product Lifecycle Management processes manage products to required profit and loss margins, customer satisfaction and quality commitments, as well as delivering new products to the market. These lifecycle processes understand the market across all key functional areas, the business environment, customer requirements and competitive offerings in order to design and manage products that succeed in their specific markets. Product Management processes and the Product Development process are two distinct process types. Product Development is predominantly a project-oriented process that develops and delivers new products to customers, as well as new features and enhancements for existing products and services.

6.1.5 Enterprise Management Process Area

The Enterprise Management Process Area includes those foundational supporting business processes that are required to run and support any large business. These generic processes focus on both the setting and achieving of strategic corporate goals and objectives, as well as providing those support services that are required throughout an Enterprise. These processes are sometimes considered to be the corporate functions and/or processes. e.g., Financial Management, Human Resources Management processes, etc… Since Enterprise Management processes are aimed at general support within the Enterprise, they may interface as needed with almost every other process in the Enterprise, be they operational, strategy, infrastructure or product processes

This process area includes those processes that manage enterprise-wide activities and needs, or have application within the enterprise as a whole. They encompass all business management processes that:

- are necessary to support the whole of the enterprise, including processes for financial management, legal management, regulatory management, process, cost and quality management, etc.;
- are responsible for setting corporate policies, strategies and directions and for providing guidelines and targets for the whole of the business, including strategy development and planning, for areas such as Enterprise Architecture, that are integral to the direction and development of the business;
- Occur throughout the enterprise, including processes for project management, performance assessments, cost assessments, etc.
- Are commonly found across multiple industries, and are managed in much the same way across most industries.

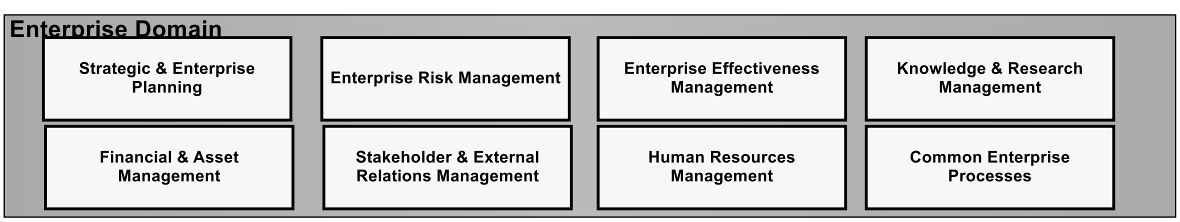

Figure 6-4 - Business Process Framework - Enterprise Management Process Areas

Many process groupings within Enterprise Management (see Figure 6-4) will contain elements that relate to both policy setting and support of the enterprise. For example, Human Resources Management is concerned with both strategy and direction as well as supporting the management of Human Resources throughout the enterprise[1].

These processes are sometimes collectively considered as the "corporate" functions and/or processes.

Common Enterprise Processes contains processes that occur throughout the enterprise. This would include the metric processes, as well as policy management processes, location management processes, and calendaring (such as appointment scheduling) when they are added, and so forth.

A key characteristic of Common Enterprise Processes is that they can execute on their own, do represent "true" business processes, but may require specialization. For example, a policy can be defined and associated with Catalog Specification to govern the content of catalogs described by the specification; or the specification of a loyalty rule may represent a specialization of the process that defines a policy rule.

6.1.5.1 Enterprise Management domain processes description

Strategic & Enterprise Planning: this enterprise management process grouping focuses on the processes required to develop the strategies and plans for the service provider enterprise. This process grouping includes the discipline of Strategic Planning that determines the business and focus of the enterprise, including which markets the enterprise will address, what financial requirements must be met, what acquisitions may enhance the enterprise's financial or market position, etc. Enterprise Planning develops and coordinates the overall plan for the business working with all key units of the enterprise. These processes drive the mission and vision of the enterprise. Enterprise Architecture Management is also a key process within this process grouping. This also directs IT across the enterprise, provides IT guidelines and policies, funding approval, etc. (note that IT development and management processes are managed within the Resource Development & Management horizontal functional process grouping).

Enterprise Risk Management: this enterprise management process grouping focuses on assuring that risks and threats to the enterprise value and/or reputation are identified, and appropriate controls are in place to minimize or eliminate the identified risks. The identified risks may be physical or logical/virtual. Successful risk management ensures that the enterprise can support its mission critical operations, processes, applications, communications in the face of a serious incidents, from security threats/violations and fraud attempts.

Enterprise Effectiveness Management: this enterprise management process grouping focuses on defining and providing the tools, methodologies and training to ensure that the enterprise operational processes and activities are managed and run efficiently and effectively. These processes ensure that the enterprise's operational processes evolve as required over time; that program and project management

[1] Note that functionality associated with a process grouping that is not required throughout the enterprise will not normally be located within Enterprise Management (for example, Human Resource Management issues specific to Call Centers are likely to be associated with the processes in Operations directly involved in this area).

processes are effective; and that quality and performance management processes are effective.

Knowledge & Research Management: this enterprise management process grouping focuses on knowledge management, technology research within the enterprise and evaluation of potential technology acquisitions.

Financial & Asset Management: this enterprise management process grouping focuses on managing the finances and assets of the enterprise. Financial Management processes include Accounts Payable, Accounts Receivable, Expense Reporting, Revenue Assurance, Payroll, Book Closings, Tax Planning and Payment etc. The Financial Management processes collects data, reports on and analyzes the results of the enterprise. They are accountable for overall management of the enterprise income statement. Asset Management processes set asset policies, track assets and manage the overall corporate balance sheet.

Stakeholder & External Relations Management*:* this enterprise management process grouping focuses on managing the enterprise's relationship with stakeholders and outside entities. Stakeholders include shareholders, employee organizations, etc. Outside entities include regulators, local community, and unions. Some of the processes within this grouping are Shareholder Relations, External Affairs, Labor Relations, and Public Relations.

Human Resources Management: this enterprise management process grouping focuses on the processes necessary for the people resources that the enterprise uses to fulfill its objectives. For example, Human Resources Management processes provide salary structures by level, coordinates performance appraisal and compensation guidelines, sets policies in relation to people management, employee benefit programs, labor relations, including Union contract negotiations, safety program development and communication, employee review policies, training programs, employee acquisition and release processes, retirement processes, resource planning and workplace operating policies. Moreover it defines the organization of the enterprise and coordinates its reorganizations.

Note that Human Resources Management processes are concerned with preparing people to carry out their assigned tasks (e.g., organizing training, remuneration, recruiting, etc.). The actual assignment of specific tasks is the responsibility of Work Force Management processes.

6.2 Business Process Framework Map

The diagram below presents a view of all domains of the Business Process Framework with their first level processes

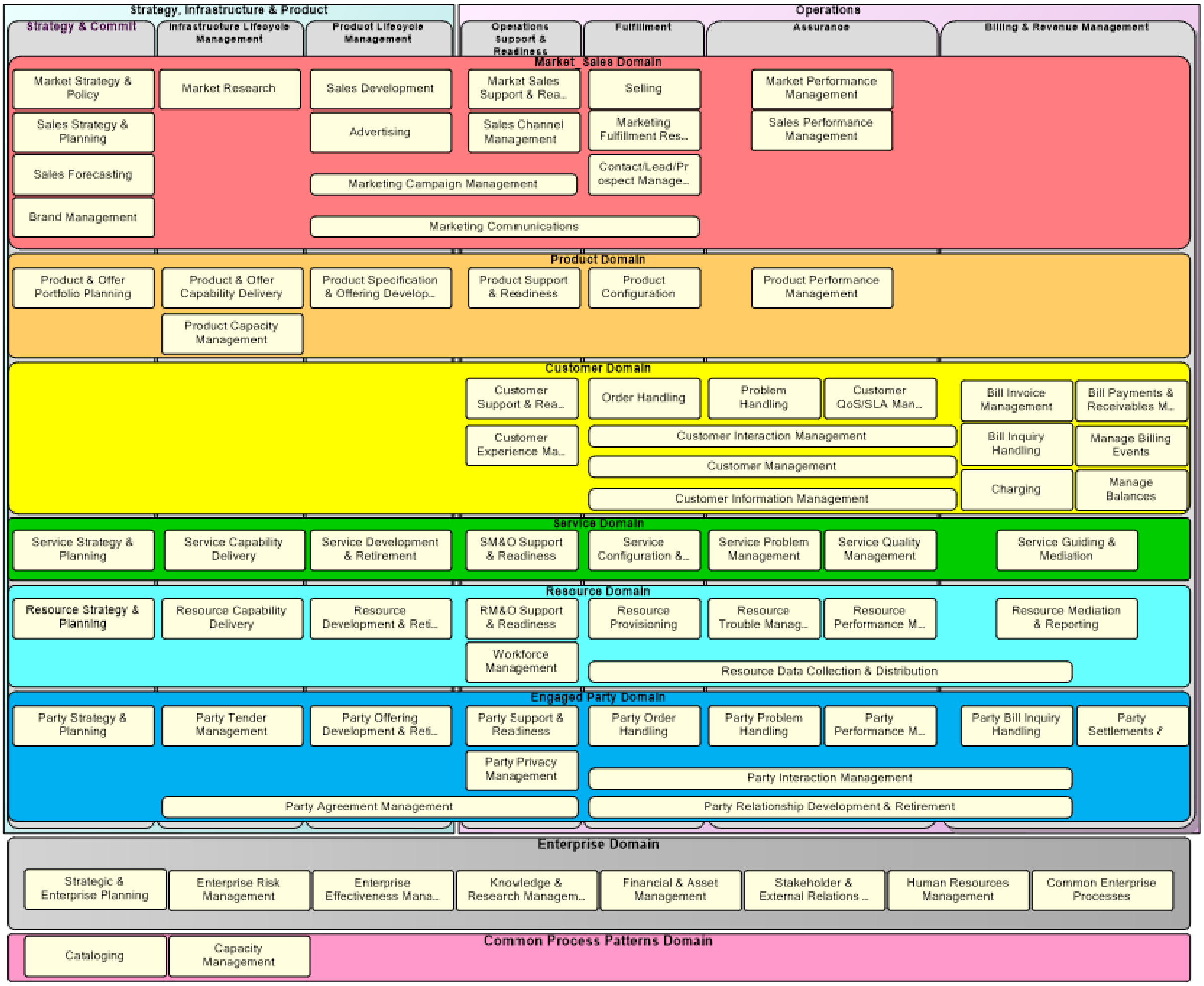

Figure 6-5 - Business Process Framework domains and processes view

6.3 External Interactions

The Business Process Framework recognizes that any single organization interacts with external parties. The major parties recognized by the Business Process Framework are customers, suppliers and partners and other engaged parties, employees, shareholders and other stakeholders.

External interactions from/to a service provider to other parties can be achieved by a variety of mechanisms including:

- Exchange of emails or faxes
- Call Centers
- Web Portals
- Business to Business (B2B) automated transactions
- Mobile Devices

- Other means….

In order to show how the Business Process Framework accommodates processes and transactions amongst a service provider and the external parties (that may be trading partners), it is useful to visualize the Business Process Framework against this external environment, and Figure 6-6 tries to illustrate this.

In Figure 6-6, the external environment is shown diagrammatically by:

- **Two horizontal "bars"**, the first one positioned above SIP and the Operations process areas (the *Sell* Side), and the second one positioned under the SIP and the Operations process areas (the *Buy* Side). These represent the two aspects of trading interactions in the external environment.
- **One vertical bar**, representing the external environment and all the external parties with links to the two horizontal bars which represent the majority on the interactions that occur.

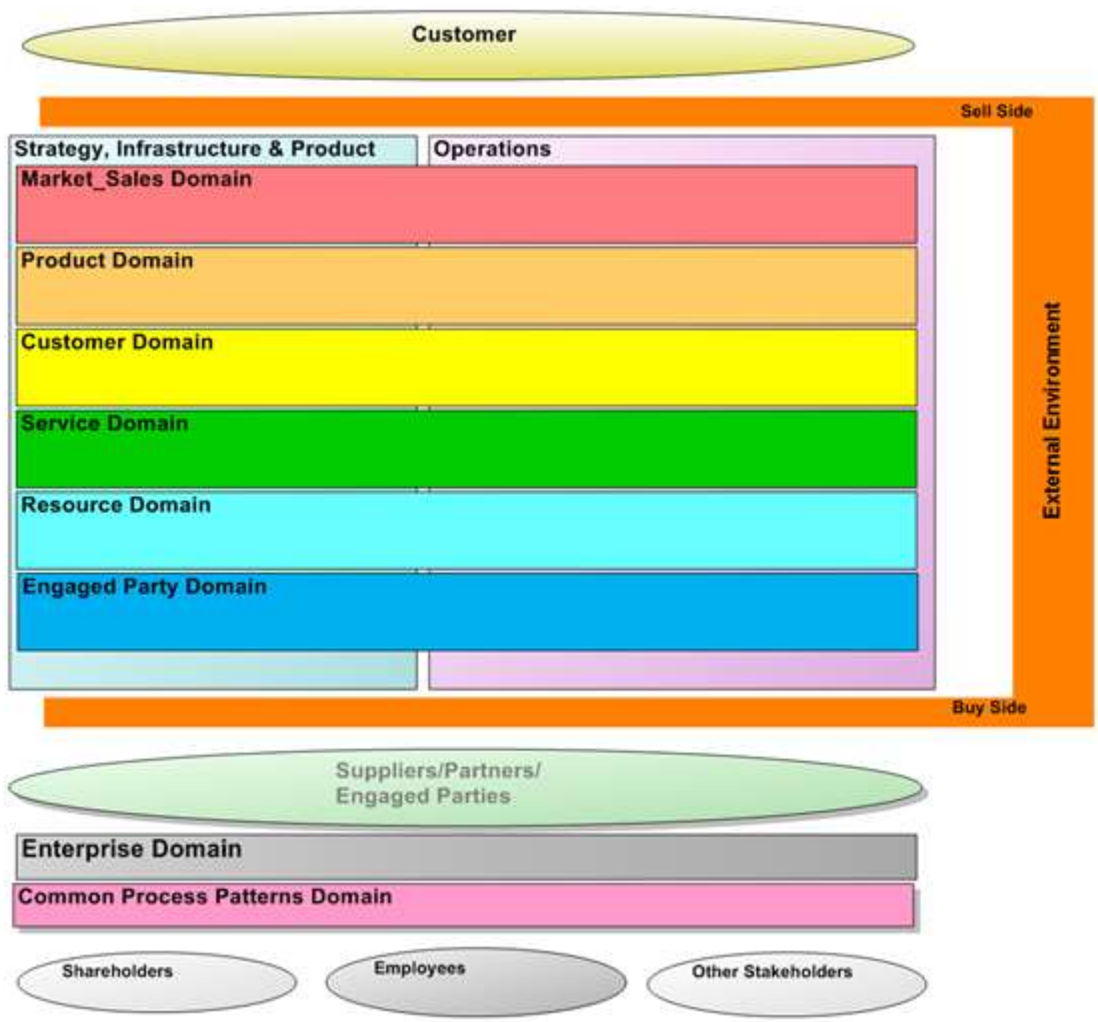

Figure 6-6 - Business Process Framework and the external environment

When the interaction with the external environment is by way of B2B trading processes, the nature and approach of these external interactions is often defined by organizations which are separate to the single enterprise. The process interactions must then be based on the concept of shared public processes, which synchronize the internal processes amongst trading partners. These shared processes have a defined "*buy*" and a "*sell*" side, which interact in a trade between a service provider and its Suppliers/Partners. Complex interactions of this kind can then be considered to consist of an appropriate set of "buy" and "sell" interactions/transactions.

B2B process interactions, and potentially other electronic interactions with customers or suppliers/partners, have specific externally specified interaction requirements. This requires that the Business Process Framework recognizes that a degree of mediation may be required as part of the process flow between the single enterprise and external parties.

When the Enterprise is trading externally, involving the use of Application-to-Application integration based on public processes, these are modeled by the added "bars". They represent the agreed industry processes to support trading with customer and partners. Some of these trading relationship with partners may involve third parties such as marketplaces, agents, trust providers, etc. which also form part of this external environment.

Further detail of the process components that support this form of external interaction is provided in GB921B and GB921C.

6.4 Process Flow Modeling Approach

A basic process flow modeling methodology has been used to show how the Business Process Framework process elements should be used to design process flows consistent with the Business Process Framework. The methodology is available in an outline form at this time and will be updated based on what proves to work well for the activity. This outline business process modeling methodology is documented separately in GB921F. There are other documents that are part of the GB921 series (Business Process Framework) which focus on various aspects regarding process flows; these are: GB921E - End-to-End Business Flows; GB921J - Joining the Business Process Framework through to Process Flows; GB921K - Construction Guidelines.

A top-down approach was adopted in the framework development phase. This enabled the definition of the Business Process Framework at the Enterprise level in a series of Level 1 process groupings. As described in the process methodology, the Business Process Framework uses hierarchical decomposition to structure the business processes.

Through hierarchical decomposition, complex entities can be structured and understood by means of the formalization of their components. Hierarchical decomposition enables detail to be defined in a structured way. Hierarchical decomposition also allows the framework to be adopted at varying levels and/or for different processes.

For the Business Process Framework, each process element has a detailed description that can include (as appropriate) the process purpose, its basic inputs and outputs, its interfaces, high level information requirements and business rules.

The Business Process Framework process flow modeling depicts process flows in a swim lane approach that drives end-to-end process and process flow-through between the customer and the supporting services, resources and supplier/partners.

Based on the above-described process modeling approach, the Business Process Framework process work starts at Level 0, the enterprise level, and shows the component Level 1 processes (see Figure 6-1). Each Level 1 process is then decomposed into its Level 2 component processes, etc.

Some examples of business process flows are presented in: GB921F and GB921E (E2E process flows).

6.5 Summary of Key Concepts

The Business Process Framework is an enterprise process framework for service providers. The processes of the enterprise fall into four major categories with twelve enterprise level process groupings in all.

The main strengths of the Business Process Framework are that it:

- Provides an enterprise-wide total business process framework for the service provider
- addresses not only operations and maintenance aspects, but covers all significant enterprise process areas
- Supports e-business, introducing concepts such as Retention and Loyalty, a new Business Relationship Context Model, Supplier/Partner Relationship Management, etc.
- Decouples lifecycle management, including development processes, from operations and day-to-day processes.
- Can represent both the Framework (static) and be used for the process flow (dynamic) view, including high level information requirements and business rules for strong linkage to automation solutions.
- Provides a process Framework reflecting the most current thinking in designing and documenting processes.
- Provides a sound reference process framework for the ICT industry in the digital services era.

The Business Process Framework (eTOM) already has this standing, not only because it builds on and enhances previous business process analysis and modeling and analysis, but because its continuing development has extensive service provider involvement, including adoption by many service providers, vendors, integrators and process tool developers.

6.6 Process Flow Concepts

The Business Process Framework includes a considerable amount of process flow modeling to support and apply the process decompositions. This modeling will continue to be developed for the process areas of the Business Process Framework which have a high priority for member organizations. Process flow modeling, definition of high level information requirements and business rules are essential elements in linking to systems analysis and design for development and delivery of automation solutions. The process decomposition and flow modeling are also critical linkages to the Frameworx systems initiatives.

This chapter addresses process flow concepts in relation to the Business Process Framework. It first gives some general information on how the process flow work is done using the Business Process Framework and then looks at the Operations processes separately from the Strategy, Infrastructure and Product processes.

6.6.1 Business Process Framework Process Flows

Process flow modeling using the Business Process Framework follows the hierarchical process decomposition and description of each process element in the hierarchies. There are two types of process flow in the Business Process Framework. First, there are the process flows for an individual process that has been decomposed to a level where it is convenient for a process 'thread' to be developed, e.g., Credit Authorization. In this context, thread is used to encompass the local process flow concerning the individual process concerned. The second type of process flow has a larger scope, and is more of a picture that connects the most important elements of several process threads to provide an 'end-to-end' process flow, e.g., service request. This type of process flow typically represents an area of business solution, and will begin to be added to the Business Process Framework in subsequent releases.

Whether a process thread or a process flow, each process involved is initiated by an event(s), e.g., a customer inquiry, and ends with a result(s), e.g., credit approved. The sequence of process steps to achieve the required overall result(s) is shown, with an association made to the high level information involved as inputs or outputs. In early input/output diagrams, each high level process showed its high level input and output, but the inputs and outputs were not defined and were not tied to a specific process activity. This deficiency is addressed in process flow modeling with the Business Process Framework, which will provide this information as more and more process flow modeling is completed.

Current process modeling methodologies use a swim lane approach to process flow diagramming, and so does the Business Process Framework. For the most part, the swim lanes are the functional layers of the Business Process Framework, e.g., CRM, SM&O, RM&O, S/PRM (replaced by the Operations part of Engaged Party) within the Operations area. Swim lanes are the horizontal layers into which the process elements and their flows are mapped. The top swim lane represents the customer. Using a swim lane approach to process flow modeling enables better:

- Process flow design, e.g., from customer request to correctly provided service
- Process flow through design, e.g., from customer to resource element

- Customer contact and interface process design, due to better visibility of the interfaces with the customer and the gaps between them
- Value add process element focus in process design
- Visibility of too many hand-offs, too much specialization, etc.

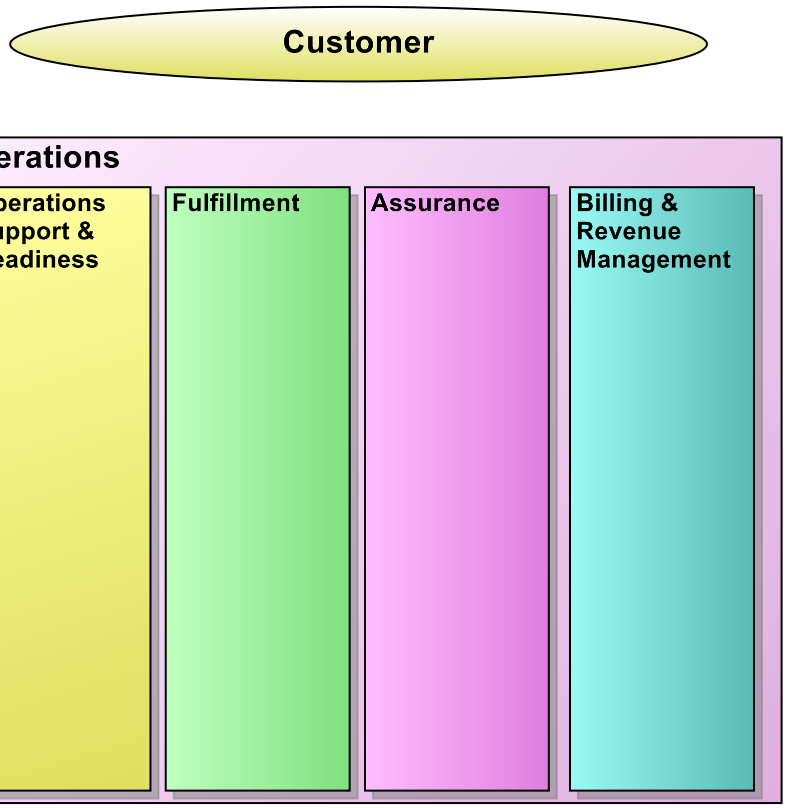

Figure 6-7 - FAB End-To-End

6.7 Forward Compatibility

The Process Framework is always a work in progress. Industry changes and shifts in the scope and behavior of Service Providers and other companies that apply the Framework mean that it must continue to evolve.

Nevertheless, great emphasis is placed on maintaining stability in the structure and content of the Framework. With such a volatile market and business environment, some change is inevitable, but this is only done – particularly at the higher levels of the Framework with a careful balance on the value of change vs. the impact on users.

An examination of the history of updates shows that continuity and stability have been achieved over a long period of releases.

PART III – INFORMATION FRAMEWORK

7 Introduction

7.1 What is the TM Forum Information Framework?

The Information Framework addresses the Digital Service Provider's need for shared information/data definitions and models. The definitions focus on business entity definitions and associated attribute definitions. A business entity is a thing of interest to the business, while its attributes are facts that further describe the entity. Together the definitions provide a business-oriented perspective of the information and data. When combined with business oriented UML class models, the definitions provide the business view of the information and data.

7.2 Purpose of the Information Framework

For many years the Business Process Framework (formerly known as the eTOM), and its predecessor the Telecom Operations Map (TOM), have provided a business process reference framework and common business process vocabulary. This framework and vocabulary have provided the communications and information industry enterprises an effective way to organize their business processes and communicate with each other.

The Information Framework (aka. SID) business view model can be viewed as a companion model to the Business Process Framework, in that it provides an information/data reference model and a common information/data vocabulary from a business entity perspective. The business view model uses the concepts of domains and aggregate business entities (or sub-domains) to categorize business entities, so as to reduce duplication and overlap. Based on data affinity concepts, the categorization scheme is necessarily layered, with each layer identifying in more detail the "things" associated with the immediate parent layer. This partitioning of the Information Framework business view model also allows distributed work teams to build out the model definitions while minimizing the flow-on impacts across the model.

Teamed with the Business Process Framework, the Information Framework provides enterprises with not only a process view of their business but also an entity view. That is to say, the Information Framework provides the definition of the 'things' that are to be affected by the business processes defined in the Business Process Framework. The Information Framework and Business Process Framework in combination offer a way to explain 'how' things are intended to fit together to meet a given business need. It should be noted that while both the Business Process Framework and the Information Framework business view model are layered, there is not necessarily a one-one relationship between the layers in each model, i.e., Business Process Framework Level 3 process elements do not only have relationships assigned to Information Framework Level 3 ABEs.

7.3 Scope of the Information Framework

The content in the Information Framework is organized using the Information Framework Model; the Information Framework was developed by the application of data affinity concepts to an enterprise's processes and data to derive a non-redundant view of the enterprise's, shared information and data. The result of this analysis is a layered framework, which partitions the shared information and data.

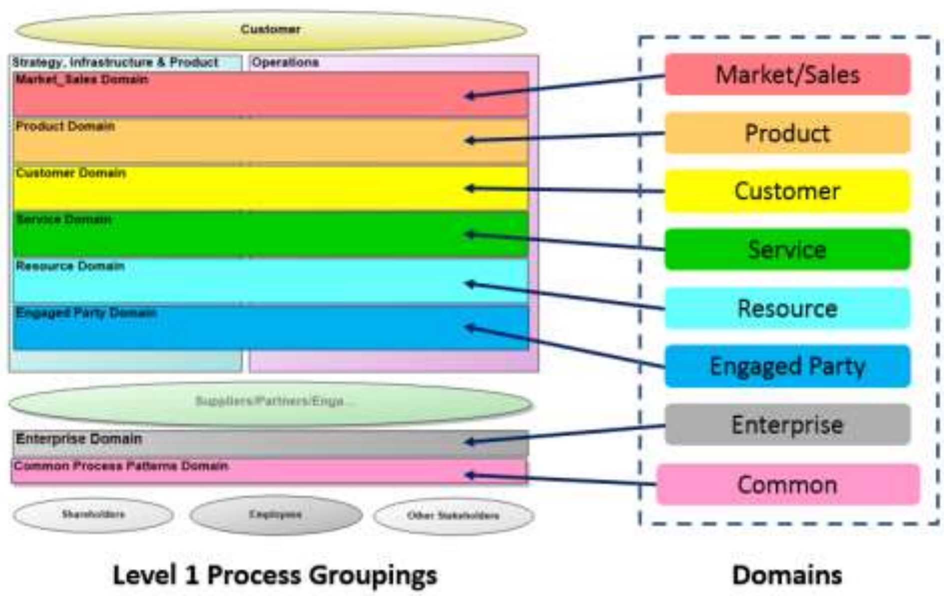

Figure 7-1 – Information Framework Domains & Business Process Framework Process Linkages

At the top layer, a set of domains is identified which are broadly aligned with the Business Process Framework business process framework as shown in Figure **7-1**. Within each domain there is a high degree of cohesion between the identified business entities, and loose coupling between different domains. This enables segmentation of the total business problem and allows resources to be focused on a particular domain of interest. It is envisioned that the use of the resultant business entity definitions within each domain, when used in conjunction with the Business Process Framework, will provide a business view of the shared information and data.

Within each domain, further partitioning of the information is achieved through the identification of Aggregate Business Entities (ABE's). Figure **7-2** shows the currently identified Level 1 ABE's. As the Information Framework business view is further expanded and defined, further partitioning of the ABE's occurs as more explicit business entities are identified.

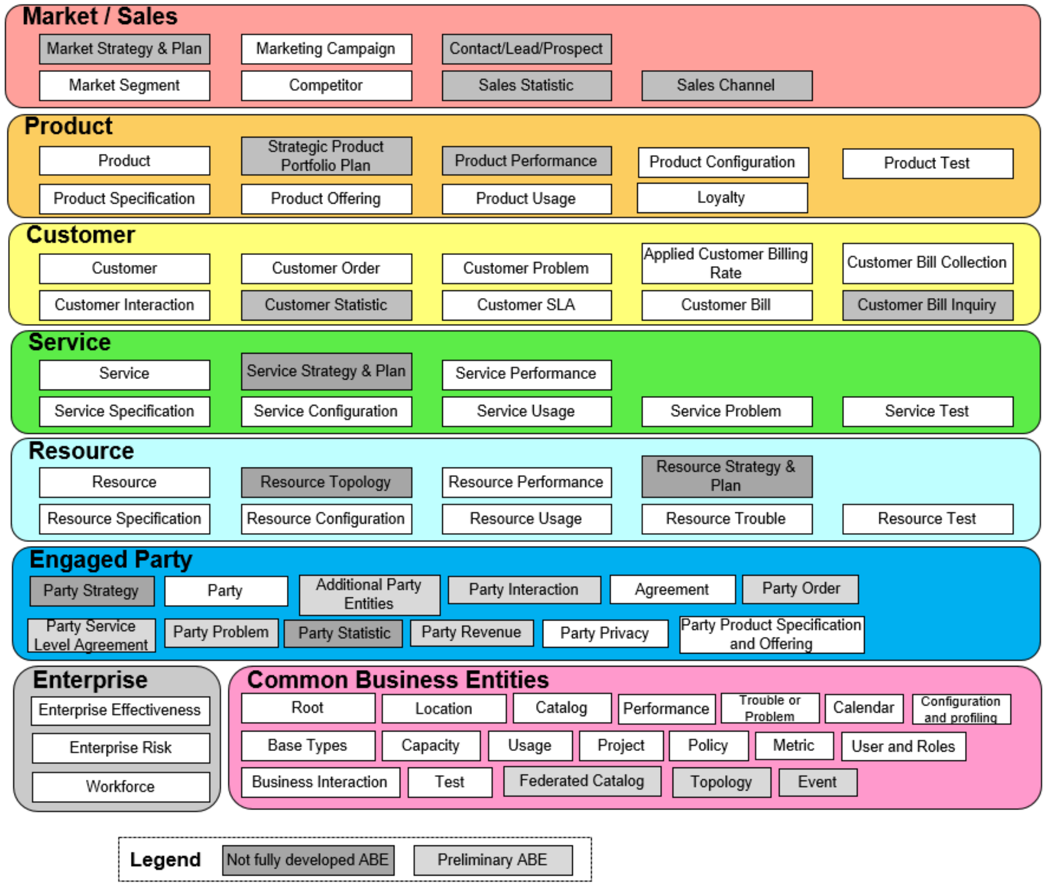

Figure 7-2 – Information Framework Domains & Level1 ABEs

The business entities along with the attributes and relationships that characterize the entities provide a view of the model that is easily understood from a business perspective. The business entities, attributes, and relationships are presented progressively developed using textual descriptions in each Information Framework addendum and in a consolidated Rational Rose UML-based model. The UML model provides an architecturally oriented business view of business entities, their attributes, and relationships to other business entities.

Domains and business entities contained in the Information Framework will expand as the Information Framework project progresses. Within each domain, TM Forum project needs were used to help scope the business entities defined and modeled in this document. The other domains and further definition of the five included domains will be presented in subsequent versions of this document or in other Information Framework documents.

The sources for the Information Framework include a variety of industry models, as well as models contributed by TM Forum member organizations. Where time permitted the contents of the Information Framework was mapped to the source models. Complete synthesis of the content of all models to find a common term for a concept was not possible. A best attempt was made to list cross-references to source models and synonyms for terms as part of the definition of the Information Framework business entities.

8 The Information Framework Concepts

8.1 Concept Definitions

8.1.1 Domain

A **Domain** is a collection of Aggregate Business Entities associated with a specific management area. Domains that make up the Information Framework are consistent with Business Process Framework level 0 concepts.

Domains are derived from an analysis of Process and Information Frameworks and have the following properties:

- Contain Business Entities that encapsulate both operations and corporate/enterprise information
- Are relatively stable collections of corporate/enterprise data and associated operations (in comparison with processes)
- Provide for robustness of corporate/enterprise data formats
- Provide clear responsibility and ownership.

8.1.2 Aggregate Business Entity

An **Aggregate Business Entity (ABE)** is a well-defined set of information and operations that characterize a highly cohesive, loosely coupled set of business entities.

8.1.3 Business Entity

A **Business Entity** represents something of interest to the business that may be tangible things (such as a Customer), active things (such as a Customer Order), or conceptual things (such as a Customer Account). Business entities are characterized by attributes and participate in relationships with other business entities. Business entity instances typically move through a well-defined life cycle.

8.1.4 Attribute

An **attribute** is a fact that describes a business entity.

8.1.5 Relationship

A **relationship** is an association of business interest between two business entities, or between a business entity and itself.

9 The Information Framework Snapshot

9.1 Introduction

This chapter describes the Information Framework that is used to organize the Information Framework model. The Information Framework provides an organizing structure in which the Information Framework business entities reside.

Figure 9-1 below shows how the domains contained within the Information Framework align with Business Process Framework level one domains/concepts. Whether taking a process or information perspective, it is important to be viewing the same set of concepts. The alignment is also a necessary enabler when mapping Business Process Framework processes to Information Framework business entities.

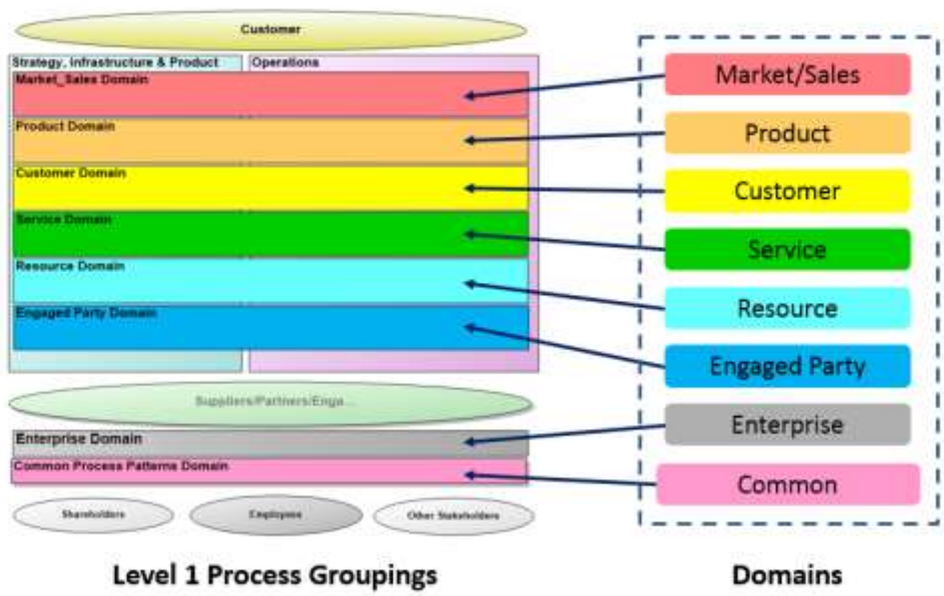

Figure 9-1- Business Process Framework/Information Framework Concepts/Domains

9.2 Information Framework – Level 1 ABEs

Figure 9-2 shows the Information Framework. The framework depicts the domains and Level 1 aggregate business entities (ABE) contained within each domain. An ABE marked as preliminary indicate that the ABE has not been fully developed, can be used if desired, but may change once its development is complete.

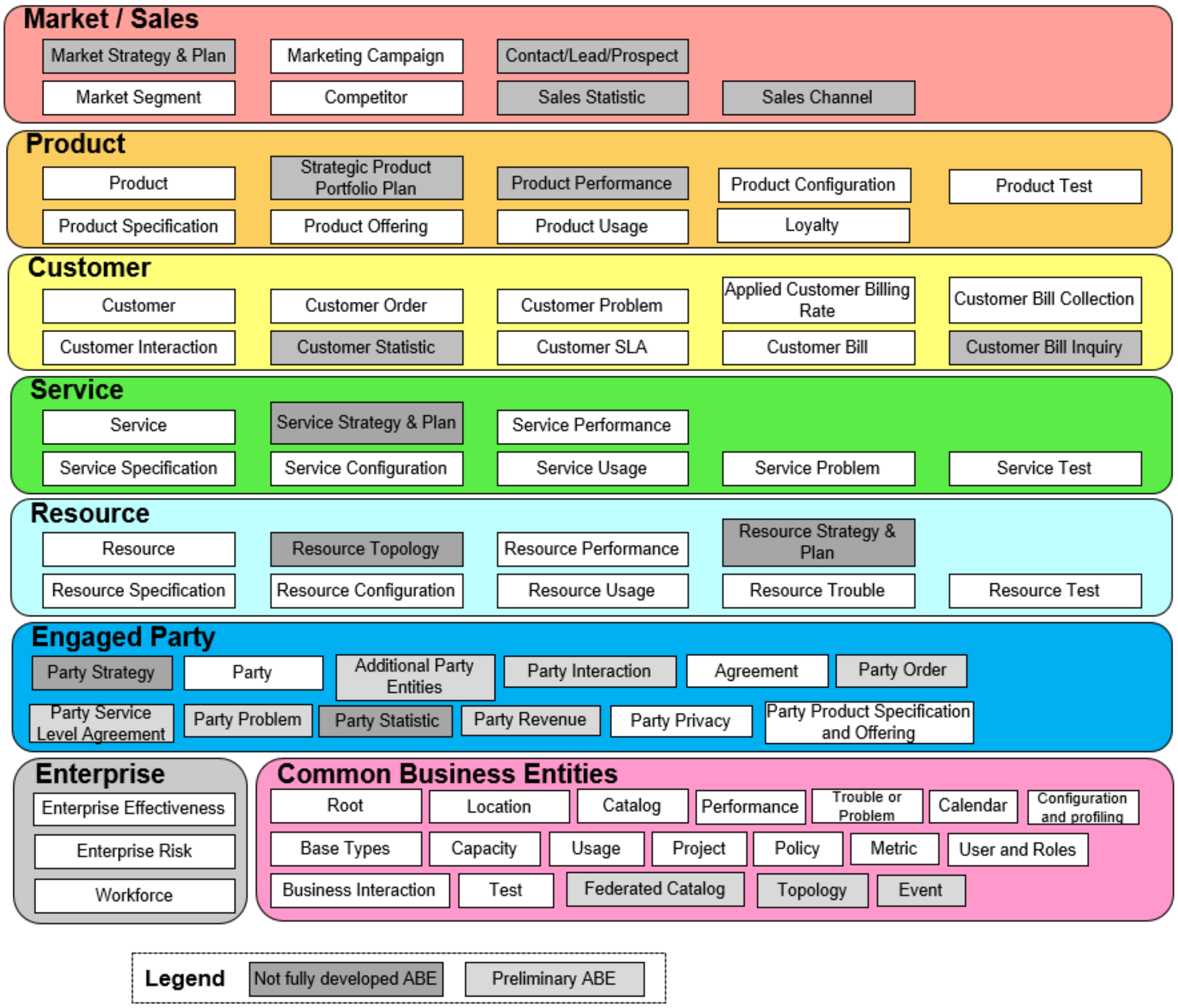

Figure 9-2 – Information Framework

As the development of the Information Framework has progressed, subsequent levels of ABEs have been identified. As an example, Figure 8-3 below shows Level 2 ABEs identified within the Market/Sales domain.

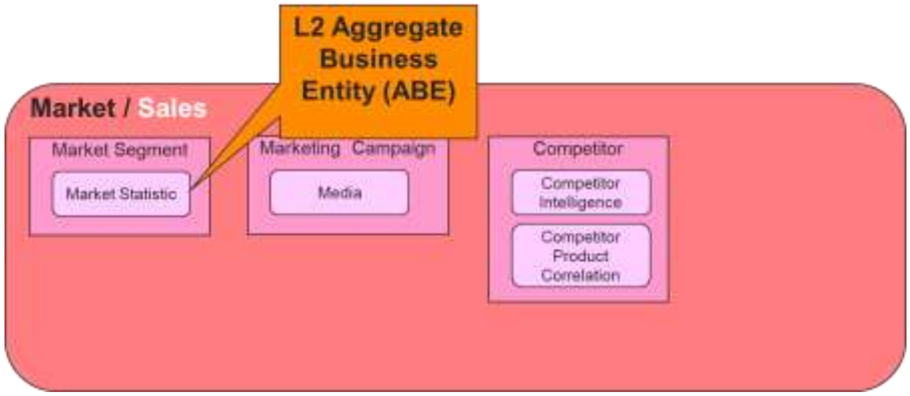

Figure 9-3 – Market/Sales Level 2 ABEs

The number of levels into which a Level 1 ABE decomposes is dependent on its complexity. Some Level 1 ABEs may decompose to two or more lower levels.

9.3 ABE Categorization

The ABE content and structure within each domain should be somewhat consistent. To ensure this, each ABE is aligned with a categorization pattern as described below. The pattern can also be used to confirm the completeness of each domain's ABE.

ABE categories include:

- Strategy and Plan
- Entity
- Entity Specification – A description of a ManagedEntity that might allow it to be built.
- Interaction – A communication with a ManagedEntity. This is a type of BusinessInteraction.
- Configuration – The internal structure of a ManagedEntity.
- Performance – The measure of ManagedEntity quality.
- Test – A means of interrogating a ManagedEntity in order to understand its state(s).
- Trouble – A problem associated with a ManagedEntity. Alarms, Outages and Faults are examples
- Price – The cost of a ManagedEntity.
- Usage – A period of time during which a ManagedEntity is in use.

Note: ABEs in Market/Sales and Common Business Domains have not yet been categorized.

	ABE Category									
	Strategy and Plan	***Entity***	***Entity Specification***	***Interaction***	***Configuration***	***Performance***	***Test***	***Trouble***	***Financial***	***Usage***
Customer		Customer	Customer SLA	Customer Interaction, Customer Bill Inquiry	Customer Order	Customer Statistic,		Customer Problem	Customer Bill Collection, Applied Customer Billing Rate	Customer Bill
Product	Strategic Product Portfolio Plan	Product	Product Specification		Product Offering	Product Performance			Product Price (Product Offering Level 2)	Product Usage Statistic
Service	Service Strategy and Plan	Service	Service Specification		Service Configuration	Service Performance	Service Test	Service Problem		Service Usage
Resource	Resource Strategy and Plan	Resource	Resource Specification		Resource Configuration, Resource Topology	Resource Performance	Resource Test	Resource Trouble		Resource Usage
Engaged Party	Party Strategy	Party, Party Privacy	Party Product Specification and Offering	Party Interaction, Party Bill Inquiry, Agreement, Party Service Level Agreement	Party Order	Party Statistic		Party Problem	Party Revenue	

Figure 9-4 – Information Framework Common Categories

Notes: 1. The absence of an ABE in a particular category means either that the domain does not exactly fit the pattern or that an ABE has not yet been identified for the category.

PART IV – APPLICATION FRAMEWORK

10 Introduction

10.1 What is the Application Framework?

The TM Forum Application Framework (a.k.a. TAM) is a functional orientated framework of reference. It describes the Service Provider ecosystem of OSS/BSS applications. The map comprises:

- **Logical function description** – Verbal specification of functionalities in the CSP ecosystem.
- **Logical functional grouping** - Common sense hierarchical grouping (or decomposition) of functionalities into applications.
- **Viewing Map** – The applications are aligned across horizontal layering of the Information Framework (a.k.a. SID) and vertical layering of the Business Process Framework (a.k.a. eTOM) so that the CSP universe is laid in a meaningful view that relates to the other frameworks.
- **Reference architecture** – The applications are related to each other through the notion of "Business Service". Each application lists the services it expose and the services it consumes yielding together a reference-architecture.

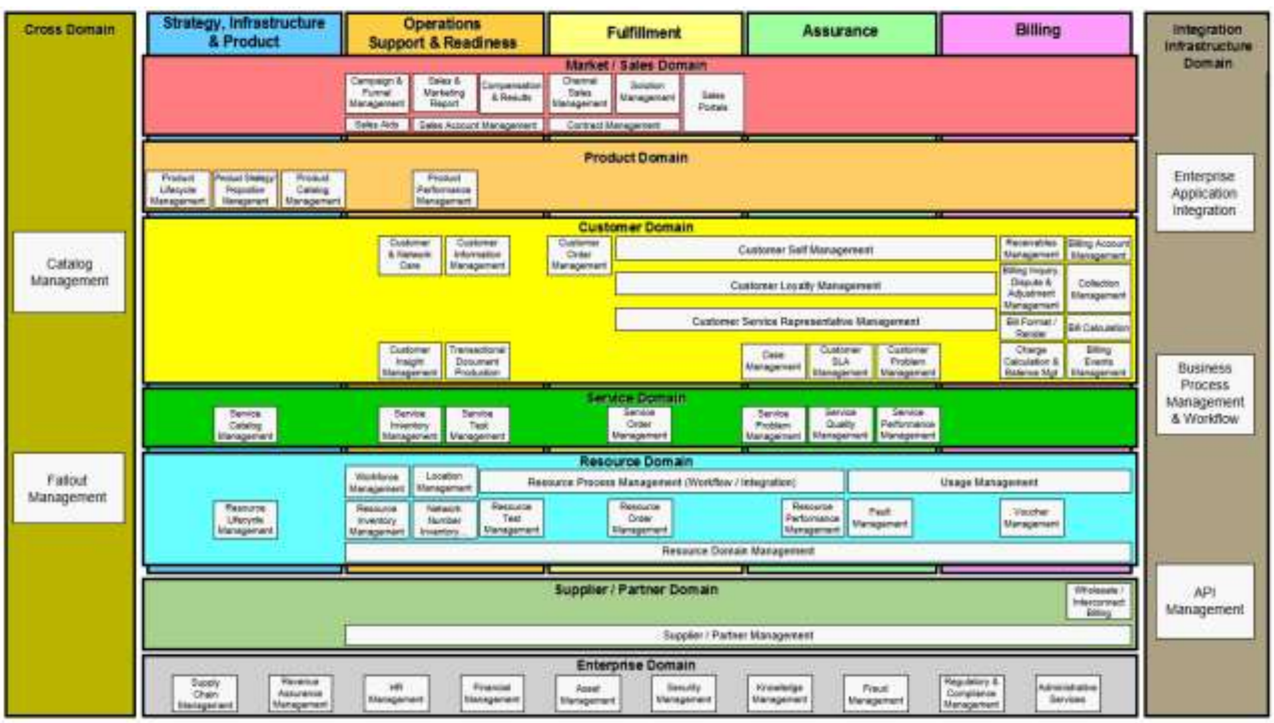

Figure 10-1- The Application Framework (TAM)

10.2 Purpose of the Application Framework

The TM Forum Application Framework is another plane in the Service Provider's ecosystem Enterprise Architecture. It provides a common reference map and language to navigate a complex systems landscape that is typically found in the CSP ecosystem. Where the Business Process Framework (a.k.a. eTOM) provides a frame of reference for *processes* and the Frameworx - Information Framework (a.k.a. SID) provides a frame of reference for *information language*, the Frameworx – Application Framework (a.k.a. TAM) provides a frame of reference for telecom *applications functionalities*.

The Application Framework provides the bridge between the Core Frameworks building blocks (Business Process Framework and Information Framework) and real, deployable, potentially procurable applications by grouping together functionalities required to support business processes in the Service Provider (SP) ecosystem.

No document like this can ever be 'right' in the sense that it represents a perfect systems infrastructure. What this document intends to give the industry is a common frame of reference that allows the various players who specify, procure, design and sell operation and business support systems to use common logical function reference architecture.

For detailed usage of the Application Frameworks please refer to document GB929U – "User Guidelines for the Application Framework".

10.3 Scope of the Application Framework

As the Business Process Framework (eTOM) addresses the business processes required to run a service provider business and the Information Framework depicts the business entities information units required for the business process, the Application Framework addresses the logical functions supporting the business processes. As such the scope of the Application Framework logical functions are focused on the management aspects of the business.

The Application Framework depicts the applications that are utilized at the enterprise. In concert with the Business Process Framework the mass of the text is focused with SIP and Operations aspects of a CSP. Since the primary benefit of the Application Framework pertains computerized systems it is more developed in the Operations area. Additional coverage of the Enterprise domain is included albeit it is of less detail.

It would be noted that the Application Framework is not covering the logical functions that are required to deliver the customer service such as network elements (this would fall under other SDOs such as 3GPP) as the TM Forum Frameworx is focused on the management aspects.

Figure **10-2** below shows how the domains contained within the Application Framework align with Business Process Framework level one domains/concepts. Whether taking a process, information or application perspective, it is important to be viewing the same set of concepts. The alignment is also a necessary enabler when mapping Business Process Framework processes to Application Framework management areas.

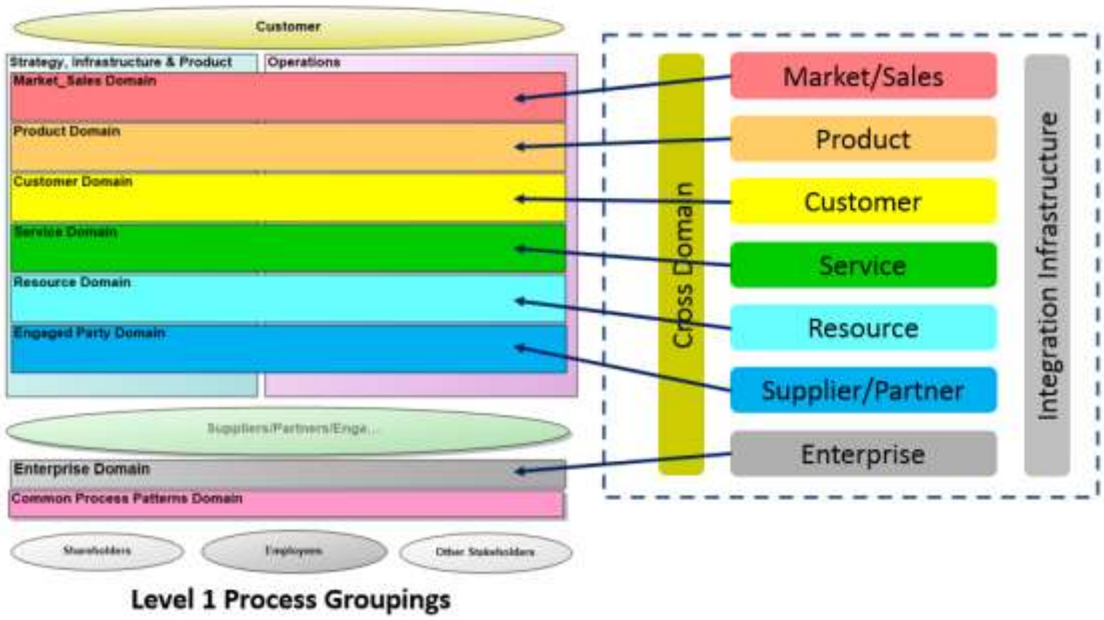

Figure 10-2- Business Process Framework/Application Framework Concepts/Domains

11 Application Framework Concepts

11.1 Application

An application is a cluster of related functionalities or other related applications. Application is an abstract logical description while a product and it capabilities is a realization of the application functionality.

For example: A word processing application is an abstract list of functions being required to author a document while Notepad, WordPerfect, MS-Word, Open office Writer, Apple iWork are just examples out of a long list of products available on the market for that purpose.

Note: The rationale of the clustering is described in a further section.

11.2 Functionality

Functionality is a verbal description of a capability required to perform a specific task.

The granularity of the functionality may vary from time to time. Coarse grain functionality may be decomposed further into lower level functionalities that can be clustered into applications.

For example: Mail merge can be considered as functionality in a word processing application. This functionality can be broken further into lower level functions such as template definitions, recipient list definition, etc.

11.3 Business Services

The notion of Business Services is outside of the scope of this document. Business Service (Formerly Known as NGOSS contract) definition is still a work in progress. The reader is referred to GB945 as a starting point. However, the notion of Business Service is important in the context of the Application Framework as the association that relates Applications to each other. Every Application provides functionality to other applications through exposed services. The functionality can be realized as a process of combining other functionalities exposed by other applications or through accessing Business Entities. It is expected that there will be at least one consuming Application of the service (otherwise the service is meaningless) and ideally only one application that exposes a particular service thus providing a normalized Application Map.

11.4 How applications are identified

11.4.1 Application definition methods

11.4.1.1 Top down

The top down approach begins with coarse grain functionality in mind and decomposing those in to fine grain functions. It is followed by an affinity analysis that clusters functions. Analysis and refinement of the resulting clusters results in a set of well-defined logical applications.

11.4.1.2 Bottom up

An approach that examines existing systems and identifies the logical functional groupings they contain. These groupings are used to amalgamate and / or refine the logical groupings of application.

11.4.1.3 Combined Approach

In practice both methods can work interleaving i.e. a system architect with optimizing an existing system or start with an abstract target architecture, decomposing/grouping functionalities back and forth, aiming towards a refined cluster of well-defined functions

11.4.2 Prime identification guidelines

Given that an application is a collection of functionalities, it makes sense to cluster functionalities based on a theme. The clustering themes of Application Framework are similar to that customarily used in SOA. There are three basic themes:

- Task centric- a collection of functionalities that are supporting a specific task; typically a business process element in Business Process Framework.

  E.g. Fallout Management

- Entity centric- a grouping based on a specific entity; typically a business entity identified by Information Framework

  E.g. Service Order management

- Utility centric- when a functionality pattern is being recognized or a functionality is repeated across the application map a generalized / normalized application is being advised

  E.g. Transactional Document Production

11.4.3 Practical identification guidelines

Practically, the Application Framework represents industry agreed reference architecture that has been evolved following market and IT trends. The following principles shaped the development of the Application Map:

11.4.3.1 Commercial availability / value

A basic measurement to identify an application can be the availability of similar products in the market. If there are available two or more products with similar functionality, it is probably an indication for a reasonable clustering of functionality thus justifying the existence of an application in the application framework. However this cannot be a prerequisite in becoming an application-framework application since it would position the application framework as a follower and not as a target reference framework. On the other hand it is expected that COTS products will be available addressing this application. Hence, an application in the framework shall have a sellable added value in the Communication Service Providers market[2]. The application framework should refrain from defining applications that has no commercial value associated with their productization.

11.4.3.2 Best IT practice

Best IT practices tend to drive architectures that affect application grouping. For example: A tiered architecture that separates backend and frontend applications would be reflected in defining the functionality of an application.

11.4.3.3 Normalization / Reusability

Whenever the same functionality is being listed as part of two distinct applications, the functionality should be extracted from all appearances into a self-contained application. The new application would expose a business service for this functionality. The original applications would consume this newly established service. The consuming applications would typically invoke this service for the purpose of providing its intended functionality. This practice is being suggested in order to streamline the Application Framework and minimize overlap. For example a customer order is viewed in a number of places in the document. An order is being captured in the various Channel Sales as well as in the Customer Order Management Application.

11.4.3.4 Self-containment

It is expected that an application will represent a coarse grained functionality collection that should in most instances be able to run on its own. I.e. it will minimize the number of consumed business services that are required to fulfill the application functionality. Note that the reusability guidance (described above) necessitates by definition the use of external services. E.g. A bill calculation application will need to rely upon external tax calculation application.

11.5 How applications are grouped

Once we have a set of fine grain applications, those are grouped based on several criteria. This section describes the various grouping options which are used throughout

[2] It may have a value in the software components market but this is out of the intention of the Application Framework.

the TAM document. The various applications are grouped according the any of these methods.

11.5.1 Invocation Context

Applications with tight invocation relationship are likely to be grouped together in a higher level application. For example: if each of the applications A and B provides business services consumed by application C it is an indication for grouping.
Another example is when shared functionality is consumed by a set of applications. For example *B0 is* sharing the same functionality in B1 B2 etc. The Bx applications are subject to grouping into a single application group B As in the common applications of Channel sales Management such as create and promote leads, Sales quotation etc. . This type of grouping would reduce the number of overall interactions between applications by creating a self-contained coarse grain application. See Figure 11-1.

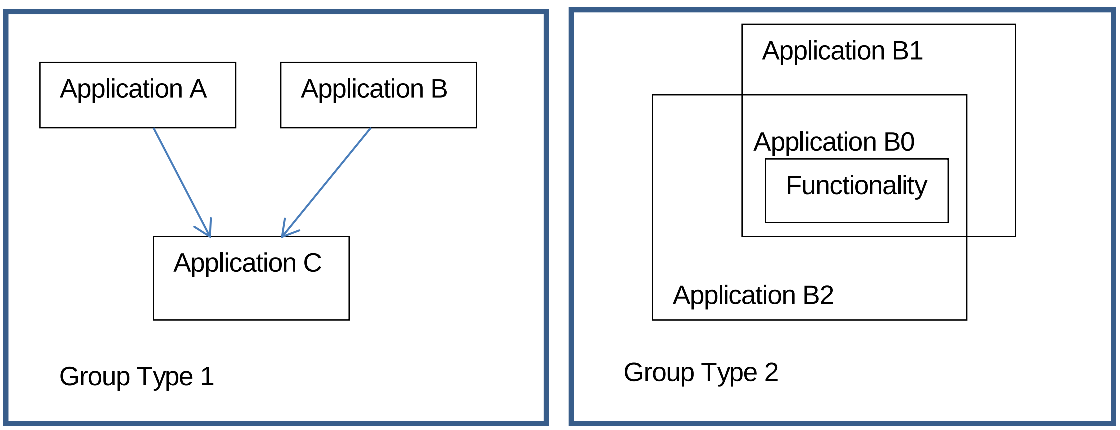

Figure 11-1 – Grouping of Applications

11.5.2 End user

The Application Framework takes into consideration the end user of the application. The end user would typically wish to see sibling functionalities required for his day to day job to be adjacent. Thus the end user using the destined product becomes an anchor for grouping the applications together.

11.5.3 Purpose

Another criterion to be used for grouping is commonality of objectives between applications. If several applications are used for sibling purpose they are likely to be grouped together.

11.5.4 Acting on a specific Information Framework entity

Applications can be grouped via the data they are acting upon. As mentioned in the document, application is automating business processes by using Information Framework entities. In some cases a group of functionalities will be related and clustered as an application based on the fact the they are strongly related to data from a specific entity with in the Information framework

11.6 How applications are mapped

One of the aspects provided with Application Framework is a viewing system. The application map approach was to reuse existing classification mechanisms from the existing frameworks (Business Process Framework & Information Framework) while extending those as necessary.

11.6.1 Embracing Process & Information Frameworks views

The Application Framework adopts the Business Process Framework vertical segments Strategy and Commit, Infrastructure Lifecycle Management, Product Lifecycle Management, Operational Support and Readiness, Fulfillment, Assurance and Billing. It also adopts the Information Framework domain horizontals, originally eight domains, the Common domain in Information Framework has been excluded from Application Framework view since it contains mostly general constructs that are used as basis for meaningful Business Entities.

The Application Framework structure maps each application in the SIP and Operation Business Process Framework domains to an intersection of both a horizontal and vertical classifications from Information Framework and Business Process Framework respectively[3].

Some applications may span across several classification segments.

11.6.2 The cross domain and its applications

Typically, an Application Framework application is listed under the domain its functionality pertains. This taxonomy, allows intra-domain (horizontal) application spanning (or grouping). However, there are some applications that span across multiple domains that needs vertical footprint. The Cross Domain Section lists applications that pertain to multiple Application Framework domains. Cross domain applications are typically created through identifying common functionality between applications across Application Framework domains (usually under the same vertical). This analysis is useful in identifying the Utility-centric business services as described in the Implementation Methodology of the TM Forum Frameworks (GB945M). A common application can be further specialized in the particular domain for the purpose of adding unique domain functionality. E. g. Catalog Management is a cross domain application and in addition in the Product, Service and Resource domains a subject specific application is described. The cross domain application concept resembles the concept of a base class and a derived class in object oriented design. Thus, a set of specialized applications (in multiple domains) that are based on the same common application can be productized in various ways. One option would be to have a unified product that encapsulates the inclusive functionality from the multiple pertaining domains. Another option would be to productize each domain functions in a separate product while encompassing the base functionality depicted in the common application. A third option would be to productize each functional grouping by its own and have the specialized application expose the common functionality by consuming the services exposed by the common application.

[3] The Supplier / Partner domain should be reworked to complete the mapping.

11.6.3 Application Integration Infrastructure

The application Integration Infrastructure is needed to complete the view of the CSPs environment, yet not reflected in other Frameworks.

This class will include applications that are not geared towards business entities (so will not be viewed in Information Framework) nor to business processes per se (so they will not have a representation in Business Process Framework).

Applications in this class will include middleware, communication application as well as business process management, workflow tools and Complex Event Processing.

Future uses of this section may expand into other aspects such as Computer Telephone Integration functionality, etc.

11.7 Naming Conventions

The Application Framework uses a standard structure in order to name of its applications. All application at any levels use a noun phrase. e.g. Customer SLA Analysis, Discounts Calculation. The use of a gerund is also acceptable e.g. Customer SLA Reporting.

Please note that a Functionality, which is a description of a capability required to perform a specific task, does not adhere to specific format as it embedded currently, but this may change as the concepts around managing Function mature over the next few releases.

11.8 Forward Compatibility

As explained earlier about the Application Framework formation, it is subject to the evolution of IT best practices. While the importance of organized evolution is closely considered, no forward compatibly restriction can be endorsed and future version of the document may detail applications that will not be compliable with the current version.

11.9 Guidelines for Function decomposition

- In the Application Framework similar application functions may be defined in different domains or applications; such definitions can be distinguished by means of enriched descriptions or reduced to a common definition for all domains.
- Splitting application functions that are too high level
- Improving names and descriptions
- Application functions shall be self-contained, connected to applications only by reference like domain, parent or origin.
- There shall be no hierarchical connections between application functions.

11.10 Definitions

Application functions are (software based) automated activities, provided as application services through human or machine Interfaces.

Application functions are computerized actions constructed to produce a result.

In a Function Model each function ideally will produce one type of result. The ambition is that Functions are defined at such granularity level that when invoked the entire outcome will be accepted.

The granularity level shall also only allow one Application Function in the Functional Model for each described function. Re-use of functions in applications will be done with references to the Function with a unique Function-Identity.

11.11 Application Functions in the Application Framework

Every application in the Application Framework consists of one or several Application Functions.

The Application's descriptions in the Application Framework will contain descriptions of functions. In addition to describing the contained functions the descriptions often puts the application in a process context and describes the business purpose.

The introduction of the layer of application functions as the lowest level of defined computerized functions has started to be called the Functional Model.

The list of functions without any hierarchy or inter-function relations is not aimed to be a model in itself, but to be the functional foundation for possible Applications, Application Models or Application Architectures.

12 Annex A

12.1 Definitions and Acronyms

Definitions and acronyms are provided here for common terms concerning Business processes and the activities occurring within them. Common terminology makes it easier for service providers to communicate with their customers, suppliers and partners.

For the Business Process Framework documentation to be understood and used effectively, it is essential that the wording listed here be interpreted using the meanings provided, rather than common usage or specific usage.

12.2 Definitions

Complementary Provider

The Complementary Provider provides additional products and services to extend the attractiveness of an enterprise's products and services and scope of its value network. Frequently, these products and services are co-branded.

Customer

The Customer buys products and services from the enterprise or receives free offers or services. A customer may be a person or a business.

Customer Operations Process

A Customer Operations Process is an end-end process that focuses totally on directly supporting customer needs, i.e., Fulfillment, Assurance or Billing. It may be initiated by the customer or be initiated by the service provider.

Digital Services

Digital services includes the Internet presence and buy and sell transaction over digital media of e-commerce. It also includes the integration of front- and back-office processes and applications to provide support and bill for the product or service. For the Business Process Framework it is even more expansive. Digital services is the integration of traditional business models and approaches with digital services opportunities.

Domain

A domain represents a unit of specific management area.

In the Business process framework it comprises a grouping of processes pertinent to a management area

Domains are derived from an analysis of Process and have the following properties:

Contain processes, that encapsulate both operations/behavior and corporate/enterprise information for the domain's business objectives

Represents a distinguishable share of the enterprise's operations

Are closely related (very cohesive) collections of corporate/enterprise processes to fulfil the domain's role in the enterprise?

e-Commerce

e-commerce is Internet presence and business buying and selling transactions over digital media.

End-to-End Process Flow

End-to-end process flow includes all sub-processes and activities and the sequence required to accomplish the goals of the process. Note that the top-level views of the Business Process Framework do NOT show end-to-end process flow since there is no indication of sequence. The Business Process Framework shows process groupings (see definition below)

The customer process flows recognized in the Business Process Framework are generic sequences of activities that need to occur in the enterprise to achieve desired results. (i.e. they are not specific to a particular ICSP Business, Product, Channel or Technology).

The Business Process Framework does not direct or constrain the way end-to-end processes can be implemented, rather it only guides the definition of standardized process elements to be used within the enterprise. In this way process elements can be assembled for a specific service provider's end-to-end process requirements. The Business Process Framework does not mandate a single way the process elements should be organized or sequenced to create end-to-end processes.

Process Grouping

The top-level view of the Business Process Framework shows process groupings. At this level of the process framework, flow is not appropriate. However, these groupings represent processes that have end-to-end results that are key measures for the enterprise.

Also termed as vertical process grouping(s).

End User

The **End User** is the actual user of the Products or Services offered by the Enterprise. The end user consumes the product or service. See also Subscriber below.

Enterprise

Enterprise is used to refer to the overall business, corporation or firm, which is using the Business Process Framework for modeling its business processes. The enterprise is responsible for delivering products and services to the customer. It is assumed that the enterprise is a "Digital i.e. Information or Communications Service Provider" (see ICSP explanation below).

Enterprise Management Process Grouping

This Process grouping involves the knowledge of Enterprise-level actions and needs, and encompasses all Business Management functionalities necessary to support the operational processes, which are critical to run a business in the competitive market. These are sometimes thought of as corporate processes and support. Some functions such as for Enterprise Risk Management (e.g. security and fraud management) have to be more tailored to Information and Communications Service Providers, but most (e.g., Financial Management, Public Relations) are not significantly different for the ICSP industry.

Party

Party, is used to mean a person, a business, technology, etc. with which a process interacts. The Customer is the most important Party. The Enterprise Management processes interact with Government, Regulators, Competitors, Media, Shareholders, the Public, Unions and Lobby groups. The Supplier and Partner Management Processes interact with Dealers, Retailers, Partners, Brokers, Third-Party Providers, Complementary Provider, Financial Provider, Service Suppliers, and Material Suppliers.

An even more generic form, entity is sometimes used instead of Party. Current business models require a greater interdependency between parties to deliver digital services in particular, and this additional complexity may impact future issues of this Framework.

Flow-through

Flow-through is automation across an interface or set of interfaces within an end-to-end process flow.

Functional Process Groupings

The functional process groupings (e.g. Customer Relationship Management, Service Management & Operations, etc.) aggregate processes involving similar knowledge. The Business Process Framework functional process groupings are the highest level decomposition of the enterprise. Functional process groupings are shown horizontally in Business Process Framework.

These functional process groupings are not hierarchical with respect to each other and are not built one above the other (i.e., one is not a decomposition of the one above), e.g., 'Service Management & Operations' is NOT a decomposition of 'Customer Relationship Management'.

Hierarchical Process Decomposition

Hierarchical Process Decomposition is the systematic approach to modeling processes above the level suitable to process flow. The Hierarchical Process Decomposition approach allows processes to be developed more modularly. See Levels below.

Information and Communications Service Provider (ICSP)

A service provider enterprise that sells information and/or communications services to other parties.

Intermediary

Within the Value Fabric, the Intermediary performs a function on behalf of the enterprise that is a part of the Enterprise's operational requirements. Intermediaries provide products and services that the enterprise either cannot provide itself or chooses not to due to cost and quality considerations. There are typically three categories of intermediaries: sales, fulfillment, and information and communication.

This term may be superseded by a more generic term related to party.

Levels

The best to way to structure a large amount of content and detail, while still allowing the higher-level views to present a summary view, is to structure the information in multiple Levels, where each Level is decomposed into greater detail at the next lower Level. This is Hierarchical Decomposition.

By having the Business Process Framework structured into multiple Levels it enables users of the framework to align their enterprise framework or their process implementations with the Business Process Framework at different levels e.g., Align at Level 1 and 2 or align at Level 1, 2 and 3.

To summarize how levels are used in the Business Process Framework.

1. The whole-of-Enterprise view (i.e., all of the Business Process Framework) is Level 0.
2. Each Horizontal (Functional) Process Grouping is Level 1.
3. All the Process Elements, e.g., Order Handling (which appear in the End-to-End Process and the Functional Process Groupings) are Level 2.
4. Level 2 Process Elements may be decomposed into Level 3 Process Elements.
5. Level 3 Process Elements may be decomposed into Level 4 Process Elements.
6. Level 4 elements are rarely decomposed into level 5 elements, but a few examples of these exist.

Offer

An offer is an aggregation or bundling of Products or Services for sale to a Customer.

Outsourcing

Outsourcing is when an enterprise contracts out one or more of its internal processes and/or functions out to an outside company. Outsourcing moves enterprise resources to an outside enterprise and keeping a retained capability to manage the relationship with the outsourced processes.

Out-tasking

Out-tasking is when an enterprise contracts with outside enterprise to provide a process, function or capability without transfer of resource. The enterprise begins using the other enterprise’s capabilities directly and electronically.

Partner

A Partner has a stronger profit and risk-sharing component in their Business Agreement with the Enterprise, than a Supplier would have. A Partner generally is more visible to the Enterprise's customer than a Supplier would be. A partner might be part of an alliance, a joint service offering, etc.

Process Flow

A *PROCESS FLOW* describes a systematic, sequenced set of functional activities that deliver a specified result. In other words, a *PROCESS FLOW* is a sequence of related activities or tasks required to deliver results or outputs.

Product

Product is what an entity (supplier) offers or provides to another entity (customer). Product may include service, processed material, software or hardware or any combination thereof. A product may be tangible (e.g. goods) or intangible (e.g. concepts) or a combination thereof. However, a product ALWAYS includes a service component.

Process Element

Process Elements can also be considered as the building blocks or components, which are used to 'assemble' end-to-end business processes. Therefore, a process element is the highest level of the constructs within the Business Process Framework, which can be used directly by the enterprise. Process elements first become visible when either a functional process grouping or a process grouping is decomposed into the second level, e.g., Order Handling,

Process elements are modular for potential reuse and independent update and/or replacement.

Resource

Resources represent physical and non-physical components used to construct Services. They are drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, and technology components. Resources may be virtualized, as well as abstract.

Service

Services are developed by a service provider for sale within Products. The same service may be included in multiple products, packaged differently, with different pricing, etc.

Service Provider (SP)

A provider or otherwise augmentor of value in a value fabric. The distinction between service provider types based on technology is fading as Service providers increasingly compete with service providers with a wildly different technology deployed. An example of this is Digital content provider Netflix transforming the Video store industry.

Subscriber

The Subscriber is responsible for concluding contracts for the services subscribed to and for paying for these services.

Supplier

Suppliers interact with the Enterprise in providing goods and services, which are assembled by the Enterprise in order to deliver its products and services to the Customer.

Supply Chain

'Supply Chain' refers to entities and processes (including those external to the Enterprise) that are used to supply goods and services needed to deliver products and services to customers.

Swim Lane

A way of depicting process flow in two dimensions by showing sequence horizontally and different actors or process types vertically. Using swim lanes to depict process flow allow for better process design in better end-to-end flow, better flow-through and better visibility of customer interactions in the process.

Third Party Service Provider

The **Third Party Service Provider** provides services to the enterprise for integration or bundling as an offer from the enterprise to the Customer. Third party service providers are part of an enterprise's seamless offer. In contrast, a complementary service provider is visible in the offer to the enterprise's customer, including having customer interaction.

TMN - Telecommunications Management Network

The Telecommunications Management Network (TMN) Model was developed to support the management requirements of PTOs (Public Telecommunication Operators) to plan, provision, install, maintain, operate and administer telecommunication networks and services. As the communications industry has evolved, use of TMN also evolved and it has influenced the way to think logically about how the business of a service provider is managed. The TMN layered model comprises horizontal business, service, and network management layers over network hardware and software resources, and vertical overlapping layers of Fault, Configuration, Accounting, Performance and Security (FCAPS) management functional areas. The latter should not be considered as strictly divided "silos" of management functions, but inter-related areas of functionality needed to manage networks and services. Indeed, ITU-T Recommendations M.3200 and M.3400 define a matrix of management services and management function sets (groups of management functions), which in turn are used to define more detailed Recommendations on specific management functions.

This influence is diminishing as the industry moves to compete in more spheres, particularly the digital services arena.

Total Enterprise Process View

The Total Enterprise Process View Includes all business processes within the Enterprise. In the Business Process Framework, the Total Enterprise Process View is also referred to as Level 0, since it includes all Level 1 process groupings.

User

See End User above.

Value Fabric

The enterprise as a participant in a value fabric is a key concept of e-business. The value fabric is the collaboration of the enterprise, its suppliers, complementary providers and intermediaries with the customer to deliver value to the customer and provide benefit to all the players in the value fabric. Digital services success and, therefore part of the definition of a value fabric, is that the value fabric works almost as a virtually integrated enterprise to serve the customer.

Vendor

Synonymous with Supplier above.

12.3 Acronyms

3GPP	3rd Generation Partnership Project
ABE	Aggregate Business Entity
B2B	Business to Business
B2B2X	Business to Business to Any Party
BE	Business Entity
BM&A	Brand Management, Market Research & Advertising
BPSS	Business Process Specification Schema
BSS	Business Support System
COTS	Commercial Off-the-shelf
CRM	Customer Relationship Management
CSP	Communications Service Provider
DRS&F	Disaster Recovery, Security and Fraud Management
E2E	End-to-end
ebXML	Electronic Business Extensible Markup Language
EDI	Electronic Data Interchange
EM	Enterprise Management
EQPIA	Enterprise Quality Management, Process & IT Planning & Architecture
eTOM	enhanced Telecom Operations Map (now Business Process Framework)
F&AM	Financial & Asset Management
FAB	Fulfillment, Assurance and Billing
FCAPS	Fault, Configuration, Accounting, Performance and Security
HTML	Hyper Text Markup Language
ICT	Information and Communications Technology
ILM	Infrastructure Lifecycle Management
IP	Internet Protocol
ISP	Internet Service Provider
IT	Information Technology
ITIL	Information Technology Infrastructure Library
KPI	Key Performance Indicator
KQI	Key Quality Indicator

M&OM	Marketing & Offer Management
NGOSS	New Generation Operations Systems and Software
NMF	Network Management Forum (predecessor of TM Forum)
OAGIS	Open Applications Group Integration Specification
OASIS	Organization for the Advancement of Structured Information Standards
OPS	Operations
OSR	Operations Support & Readiness
OSS	Operations Support System
PaDIOM	Partner, Design, Integrate, Operate and Monetize
PIP	Partner Interface Process
PLM	Product Lifecycle Management
QoS	Quality of Service
R&DTA	Resource & Development, Technology Acquisition
RD&M	Resource Development & Management
RFP	Request for Proposal
RM&O	Resource Management & Operations
RNIF	RosettaNet Implementation Framework
RSA	IBM's Rational Software Architect
S&EP	Strategic & Enterprise Planning
S&ER	Stakeholder & External Relations
S/P	Supplier/Partner
S/PRM	Supplier/Partner Relationship Management
SC	Strategy & Commit
SCD&M	Supply Chain Development & Management
SD&M	Service Development & Management
SDO	Standards Development Organization
SID	Shared Information & Data Model
SIP	Strategy, Infrastructure and Product
SLA	Service Level Agreement
SM&O	Service Management & Operations
SOA	Service Oriented Architecture
SP	Service Provider (see also ICSP)
TAM	Telecom Application Map (now Application Framework)
TM Forum	TM Forum (Transformation and Monetization Forum) (see also TMF)
TMF	TM Forum (Transformation and Monetization Forum) (see also TM Forum)
TMN	Telecommunications Management Network
TOM	Telecom Operations Map
UML	Unified Modeling Language

UN/CEFACT	United Nations Center for Trade Facilitation and Electronic Business
VC-MC	Value Chain Market Center
W3C	World Wide Web Consortium
XML	Extensible Markup Language
XSD	XML Schema Definition

13 Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

13.1 About this document

This is a TM Forum Guidebook. The guidebook format is used when:

- The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.
- Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.
- Information about the marketplace is provided, as in the report on the size of the OSS market.

13.2 Document History

13.2.1 Version History

Version Number	Date Modified	Modified by:	Description of changes
15.5.0	Nov 2015	Frameworx Team (Editor: Alfred J. Anaya)	Initial issue Ongoing merging of Frameworx Concepts & Principles documents from eTOM, SID and TAM
15.5.1	Nov 2015	Alicja Kawecki	Updated cover, minor formatting/cosmetic fixes prior to publishing
15.5.2	Apr 2016	Alfred Anaya	Updated corrected eTOM L2 view with missing processes in the resource domain. Updated slightly the Acronyms section.
16.0 (DRAFT)	Apr 2016	Alfred Anaya	Inserted SID/eTOM mapping section and updated fields and

			related information accordingly
16.0.1	8 Jun 2016	Alicja Kawecki	Updated cover, header; minor cosmetic edits prior to publication for Fx16
16.5.0	Nov 2016	Avi Talmor	Updated for Rel 16.5
16.5.1	23 Nov 2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16.5
16.5.2	23 Feb 2017	Alicja Kawecki	Updated cover, header, footer and Notice to reflect TM Forum Approved status; applied rebranding

13.2.2 Release History

Release Number	**Date Modified**	**Modified by:**	**Description of changes**
15.5.0	Nov 2015	Frameworx Team (Editor: Alfred J. Anaya)	Initial release of document Ongoing merging of Frameworx Concepts & Principles documents from eTOM, SID and TAM
16.0.0	Apr 2016	Frameworx Team (Editor: Alfred J. Anaya)	Inserted SID/eTOM mapping section and updated fields and related information accordingly
16.5.0	Nov 2016	Avi Talmor	Updated for Rel 16.5

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